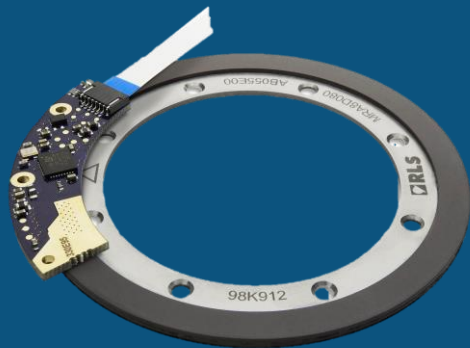


Chapter 03: Proprioceptive (Internal) Sensors

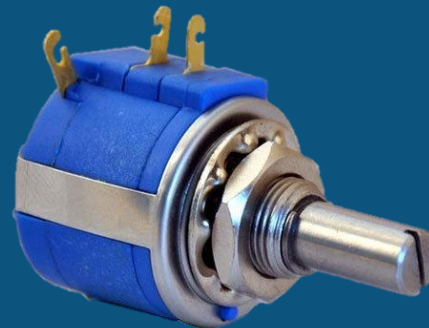
Robotics III: Sensors and Perception in Robotics

Tamim Asfour

<http://www.humanoids.kit.edu>



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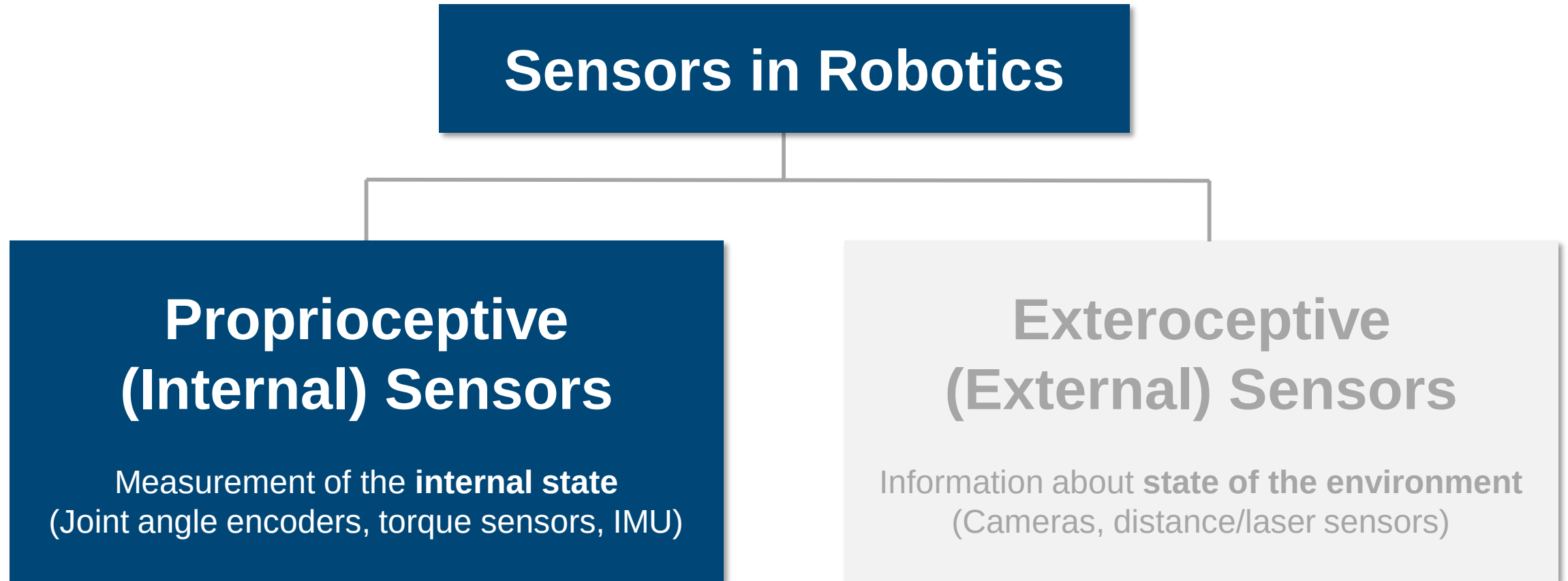


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Proprioceptive and Exteroceptive Sensors



Outline

- Proprioceptive (Internal) Sensors
 - Position sensors
 - Force sensors
 - Torque sensors
 - Inertial sensors

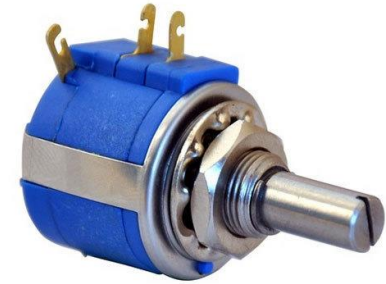
Internal Sensors – Examples

■ Position sensors

- Optical encoders
- Magnetic encoders
- Potentiometers
- Draw wire sensors



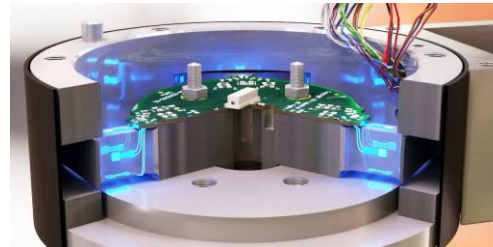
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■ Force sensors

- 1D force sensors
- 6D force/torque sensors



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■ Torque sensors

- Analog
- Digital



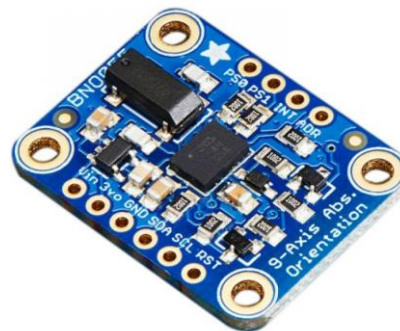
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■ Inertial sensors

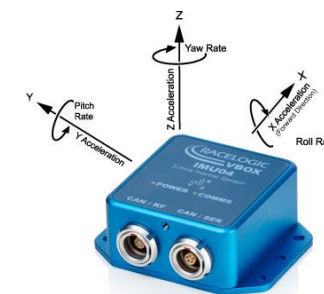
- Accelerometers
- Gyroscopes

■ Integrated attitude sensors

- IMUs
- AHRS

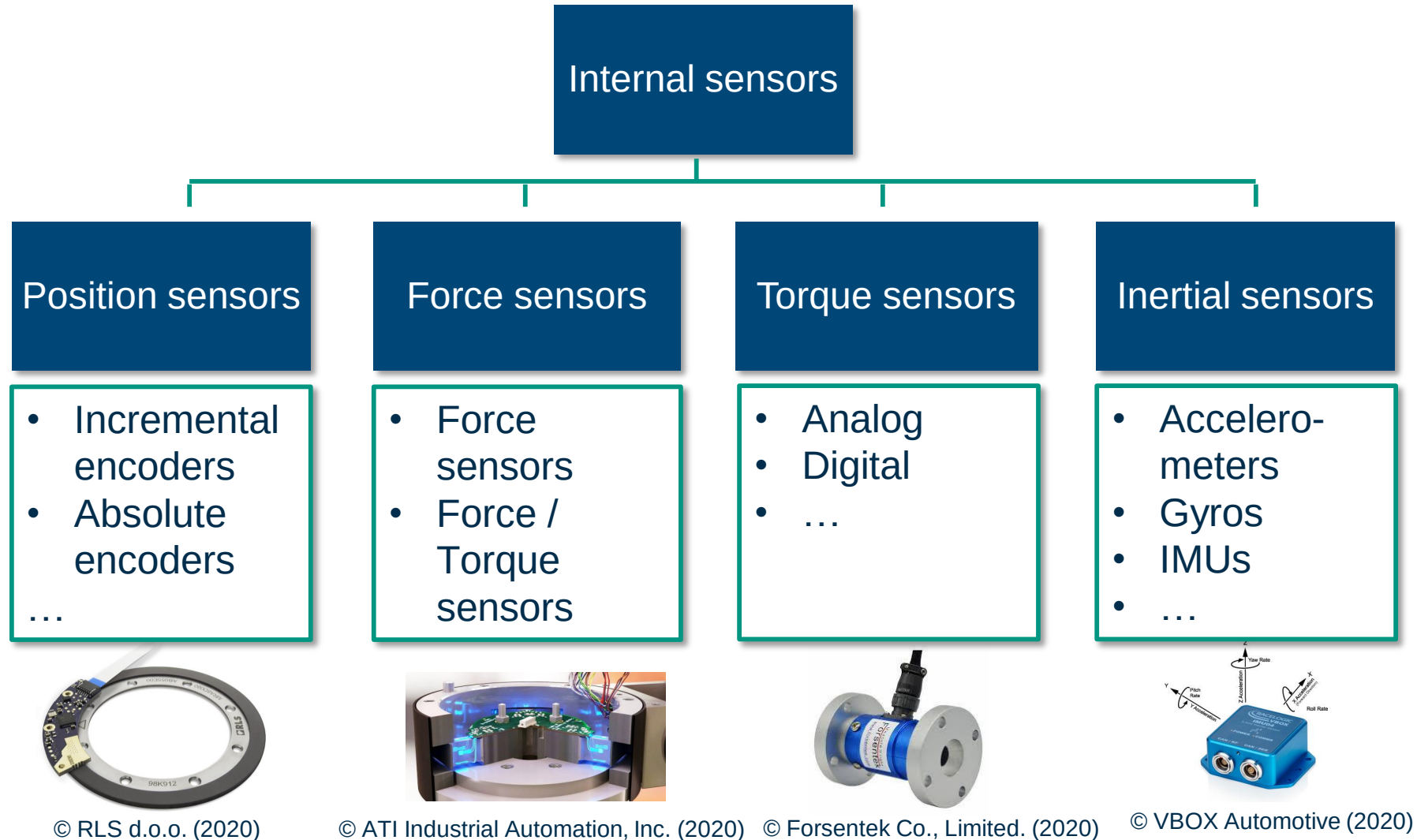


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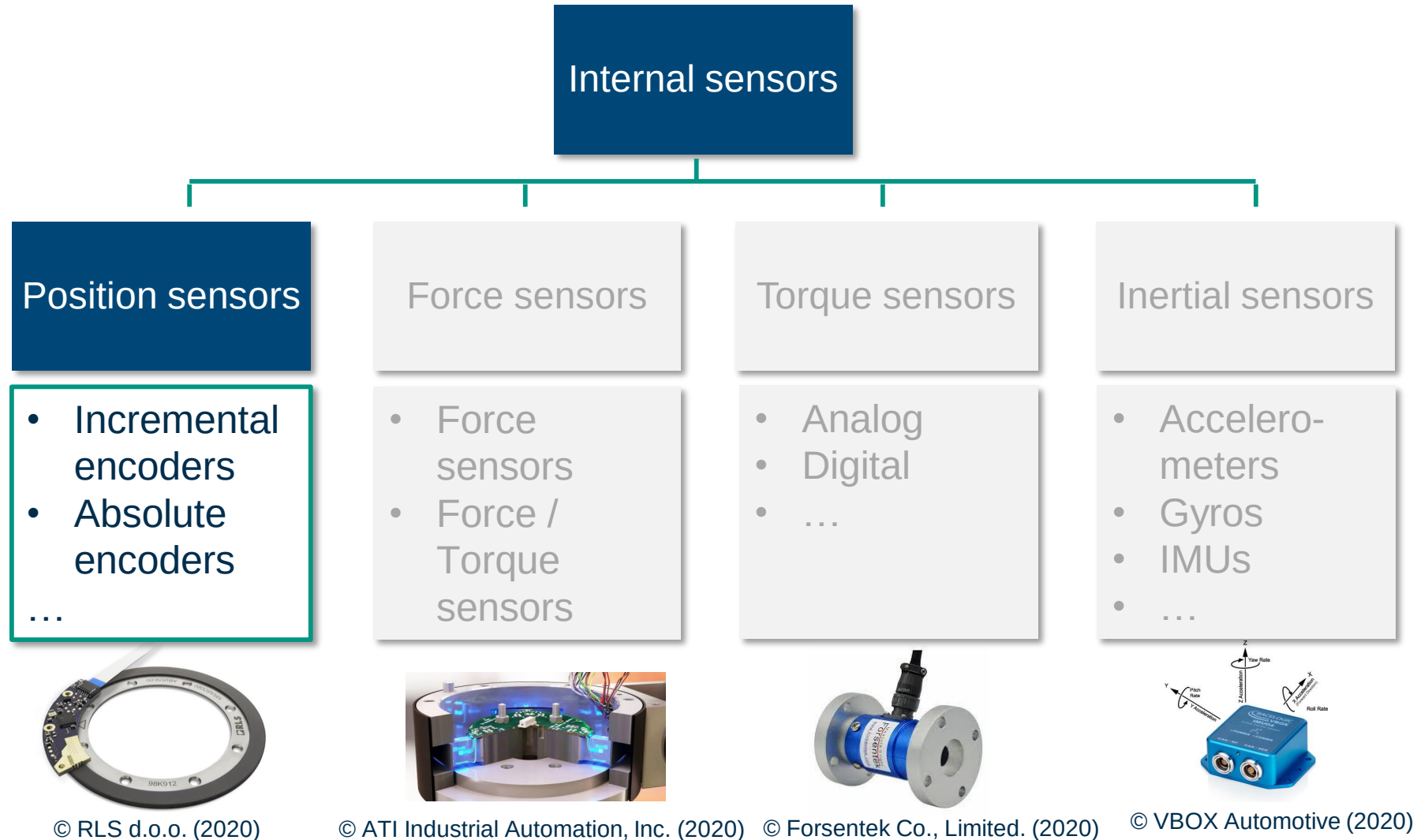


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Internal Sensors – Overview

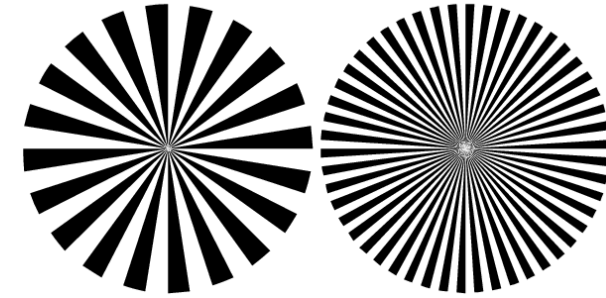


Internal Sensors – Overview



Position Sensors

- Internal position sensors measure the joint displacements of the robot:
 - **Rotary joints** → **Angular sensors**
 - **Prismatic joints** → **Distance sensors**
- Three most widely spread sensor technologies for rotary position encoders are:
 - Optical
 - Magnetic
 - Potentiometers
- One can differentiate between different rotary encoders by the way they are mounted:
 - On-axis
 - Off-axis



Optical encoder discs

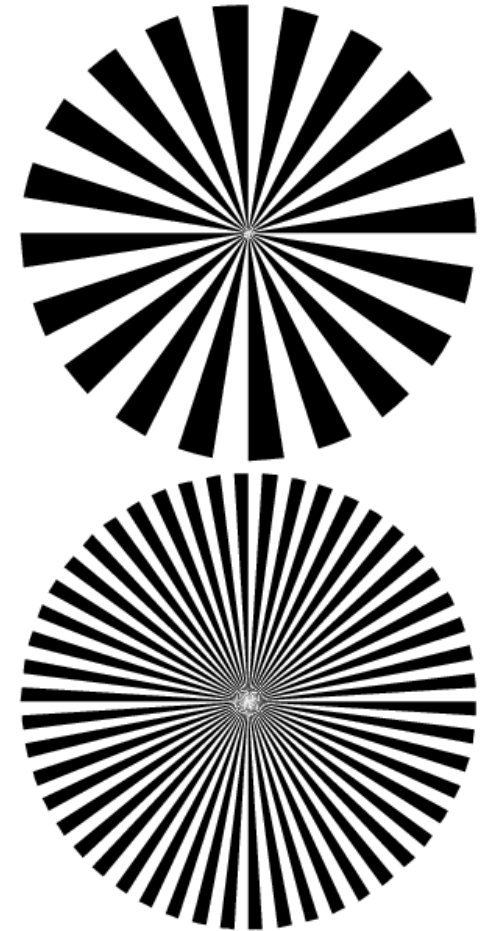


Magnetic off-axis position sensor

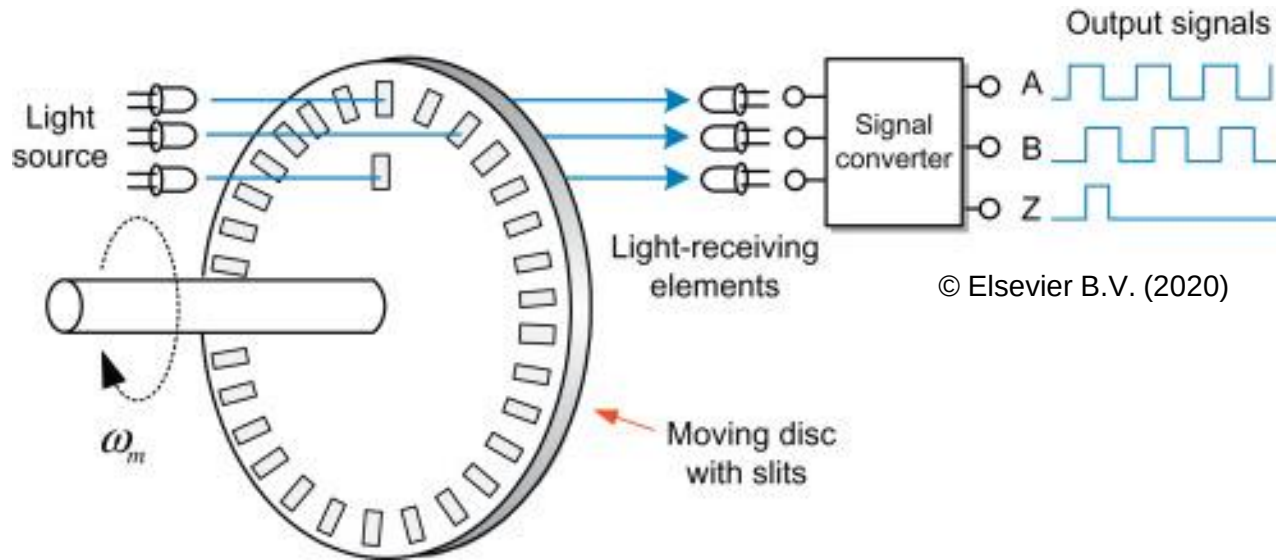
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Optical Encoders

- Sense the angular or translational displacement of a robotic joint using optical measurement methods
- **Principles of operation:**
 - Sending light through a partially transparent encoder disc
 - Measure reflections of a partially reflective disc
 - Correlation between successive camera images (optical flow, e.g. in a computer mouse)
- **Types:**
 - **Incremental encoder:** Only changes in the position (increments) are detected. The absolute position cannot directly be obtained.
 - **Absolute encoder:** The sensor can measure the absolute position at any given time



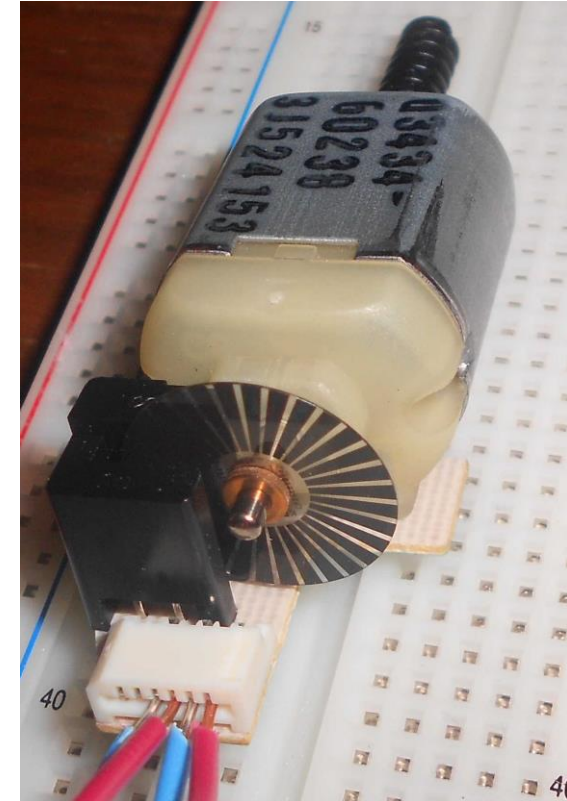
Incremental Optical Encoders



Principle: **Partially Transparent encoder disc**

- Light source shines on detector
- Partially transparent encoder disc periodically interrupts the light beam
- The interruptions are detected and summed up to form the position values

Reflective encoders count the reflected light impulses (disk not transparent); otherwise, similar operation



Incremental optical encoder with partially transparent disc attached to a small DC motor

Single-track Optical Encoders (Angle)

Only one code track and one signal line

■ Advantages

- Very simple to construct
- Cheap
- Only one light source (LED)
- Only one detector (phototransistor)
- Only one signal line needs to be processed

■ Problem

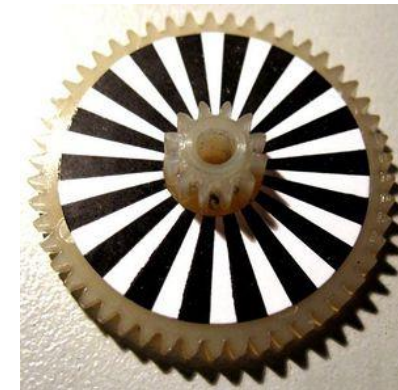
- Direction of rotation remains unknown



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© Arexx Engineering (2020)



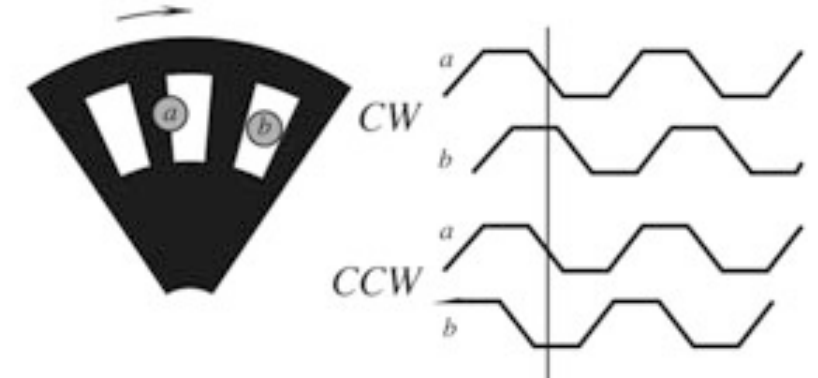
Single-track encoder disc of the ASURO educational robot

Quadrature Encoders (Angle and Direction)

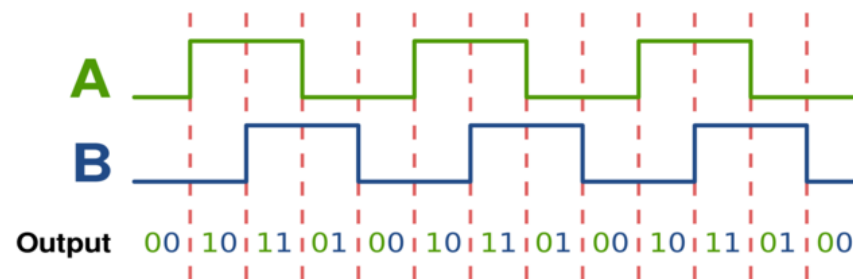
- A second, phase-shifted code track enables detection of the direction of rotation
- Phase-shift is typically 90°

Operation principle *quadrature encoder*:

- At every signal edge, the polarity of that edge (rising, falling) and the state of the other signal (high, low) are sampled
- This information encodes the direction

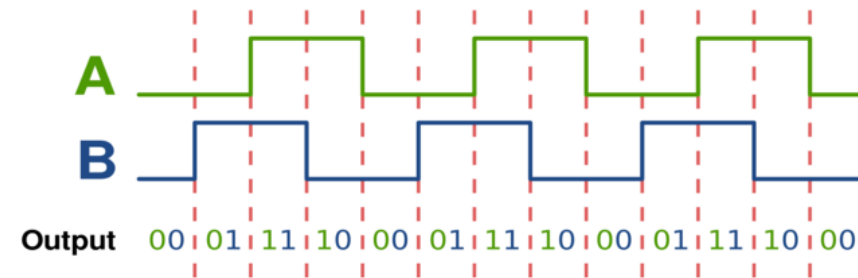


From Fraden J.: Handbook of Modern Sensors



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Forward rotation



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Backward rotation

Triple-track Optical Encoders (Angle, Direction, Initial Position)

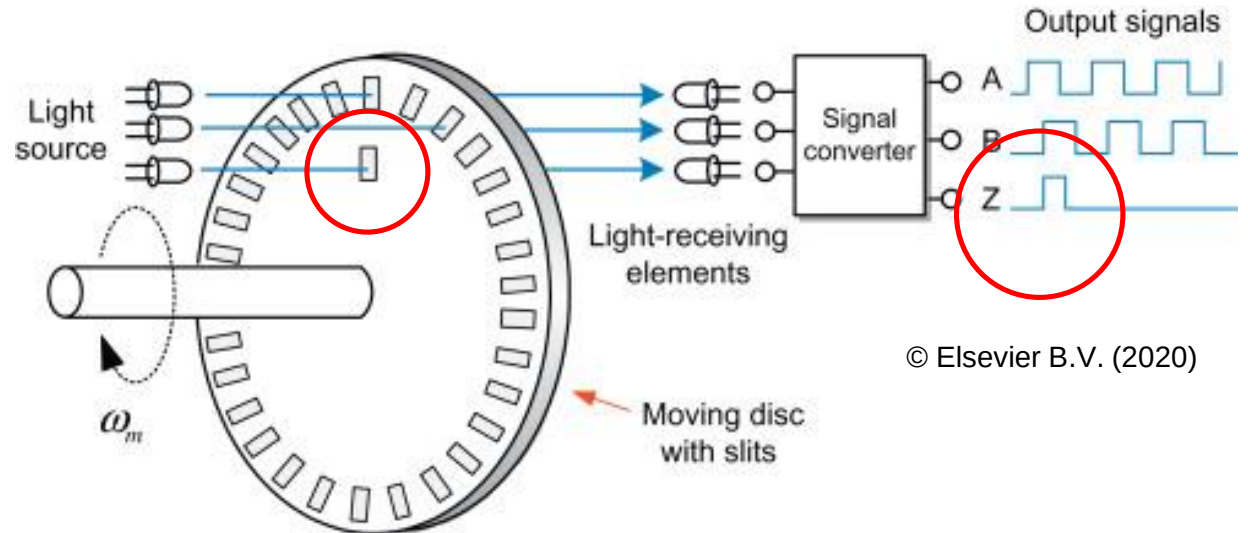
Fundamental **problem** with incremental encoders: The **initial position is unknown**

Solution:

- Adding a third track and a third light source/detector pair to encode an initial position
- This enables finding a defined initial pose within one rotation

Still a Problem:

- In case that the range of motion is larger than one rotation, the initial position cannot be encoded



Magnetic Incremental Encoders

- The encoder disc consists of a magnetized ring with **alternating polarity**
- A sensor IC detects the changes in polarity as the disc rotates
- The IC comprises two magnetic field sensors that produce a **phase-shifted quadrature signal**
- The most common sensor principle uses the **Hall-effect**
- **Cheaper** and **more robust** than optical encoders for medium resolutions



© ams AG (2020)

Magnetic encoder disk with quadrature encoder IC

Absolute Encoders

Homing with Incremental Encoders

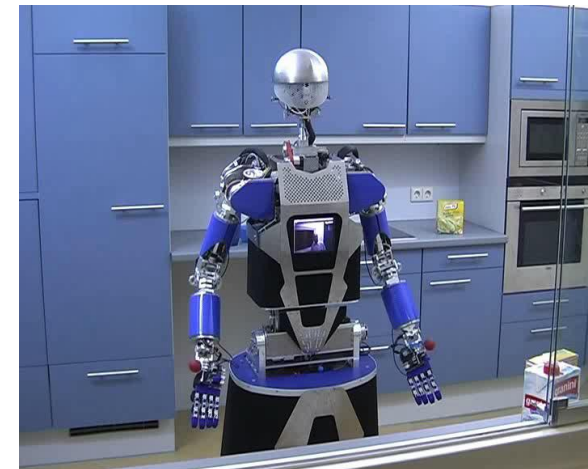
- A system with incremental encoders initially does not know in which configuration its joints are (information is lost at power-off)
- After start-up, known initial positions need to be reached
 - Marked by third encoder track
 - Marked by limit switches
 - Marked by mechanical hard-stop
- From there any desired initial position can be reached → **Homing**



© Omron Corporation (2020)



Homing of a USB conference cam
after power-on



Homing of ARMAR-IIIa

Absolute Encoders

Measure the **absolute position** of a joint **at any time**

- No information is lost due to power-off
- No need for a homing procedure
 - Faster readiness for action of the entire robotic system!
- Interface not only transmits signal edges („ticks“) but the entire position information

Disadvantages compared to incremental encoders:

- More complex mechatronics
- Higher data throughput, more complex protocols
- More expensive



Magnetic off-axis position sensor

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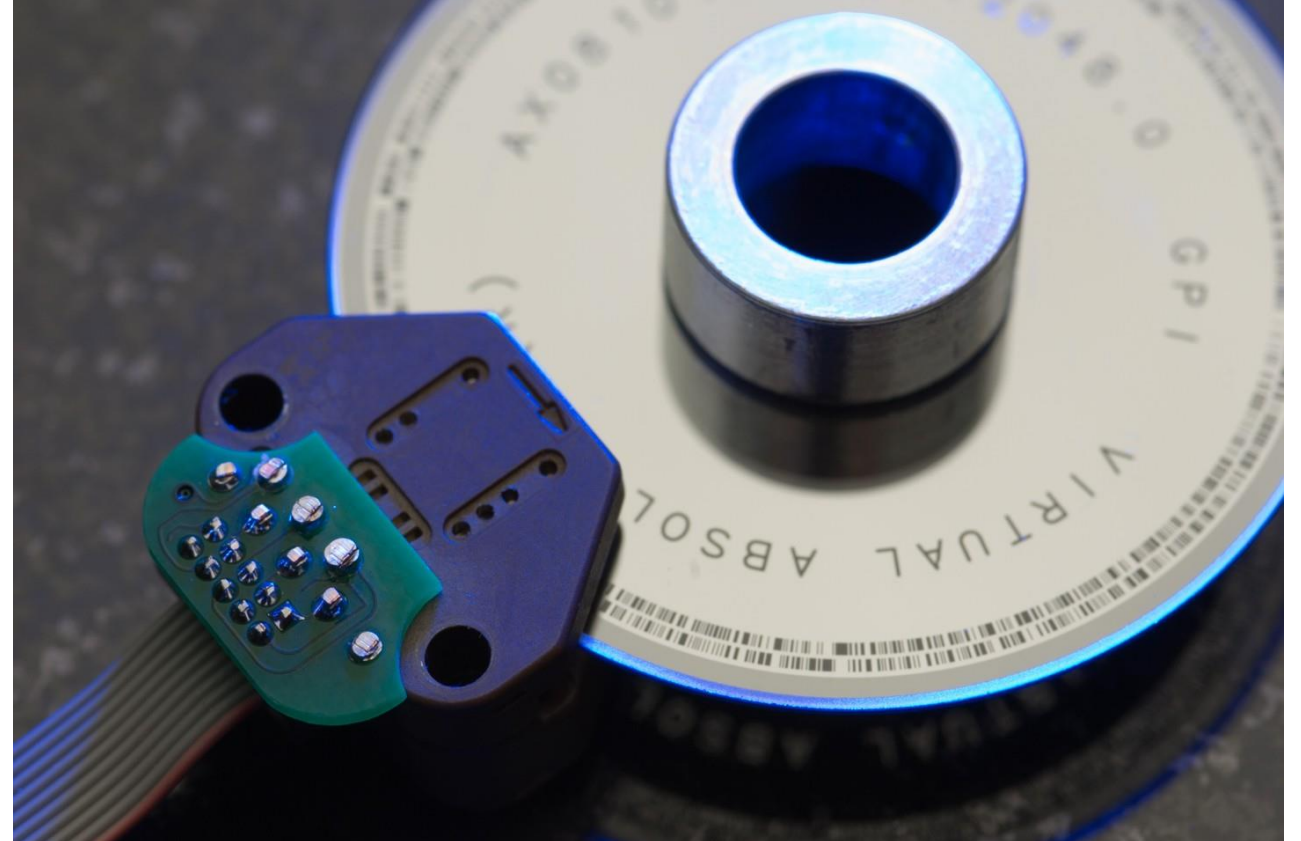


Encapsulated optical on-axis absolute encoder

© WayCon Positionsmesstechnik GmbH (2020)

Optical Absolute Encoders

- A light source shines through a structured encoder disc
- Each position along the encoder disc creates a unique light pattern on the detector array
- The light pattern is converted into a position value and transmitted via a digital interface



© Gurley Precision Instruments, Inc. (2020)

Codes for Optical Absolute Encoders

Binary code

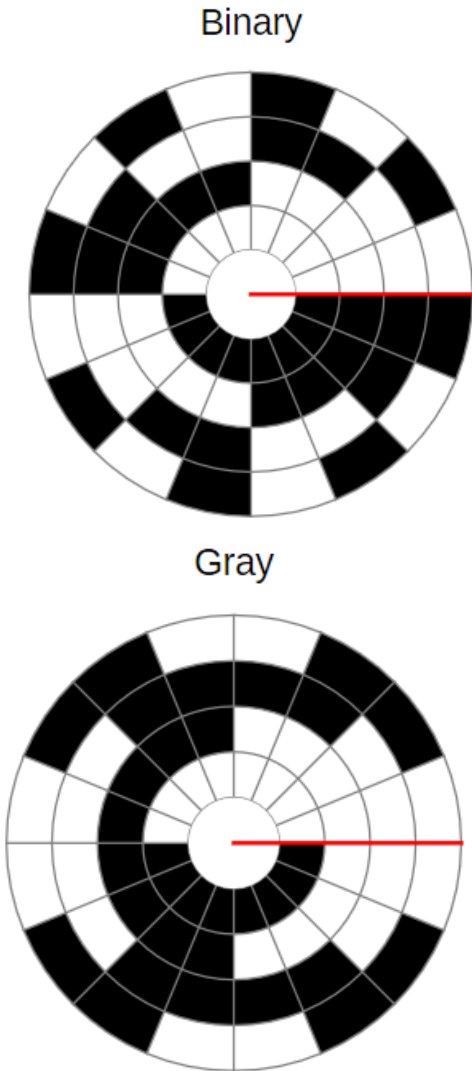
- Transparent patches at a given position are interpreted as logical ones
- Position is encoded as binary number
- **Problem**: During the transition between two positions, two bits might not change at the exact same time, causing **erroneous transient states**

Gray code

- In Gray-code discs **only one bit changes between two adjacent positions**
- Transient states do not occur

Interactive visualisation with erroneous transient states:

<https://demonstrations.wolfram.com/GrayCodesErrorReductionWithEncoders/>



Taken from Wolfram Demonstrations,
© Wolfram Demonstration Project & Contributors (2020)

Codes for optical absolute encoders

■ Binary code

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Bit 1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
Bit 2	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
Bit 3	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
Bit 4	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

■ Gray code

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Bit 1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
Bit 2	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
Bit 3	0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0
Bit 4	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0

Magnetic Absolute Encoders I

On-axis encoders

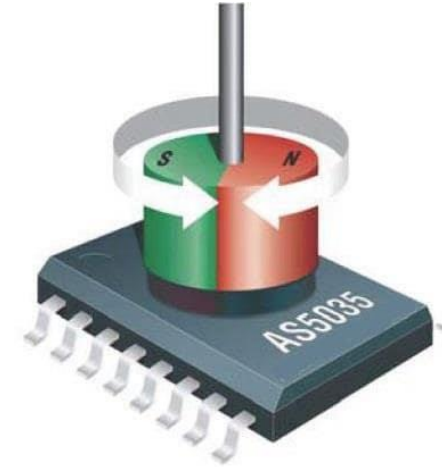
- A single magnet rotates with the shaft
- The sensor IC is fixed, positioned in very close proximity to the magnet
- An array of Hall-sensors within the IC detects the orientation of the magnetic field and computes the position

Advantages

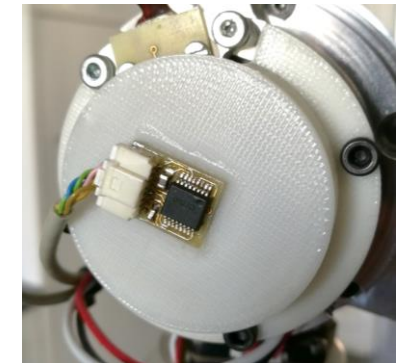
- Mechanically simple
- Relatively cheap

Disadvantages

- Not well suited for hollow shafts
- High demands on distance and centricity



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Magnetic on-axis absolute encoder IC on the shoulder of ARMAR-4

Magnetic Absolute Encoders II

Off-axis encoders

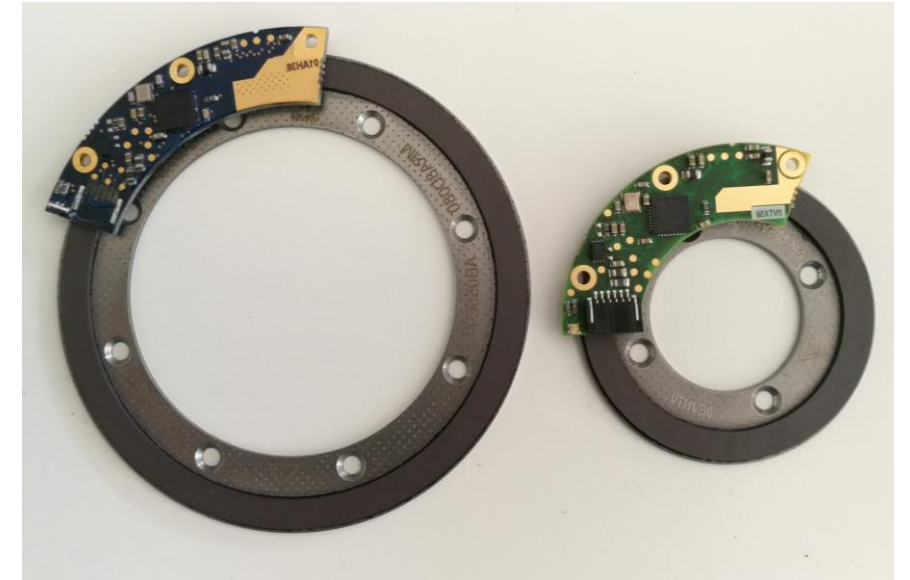
- A magnetic code ring with complex magnetization rotates with the shaft
- Each position (within the resolution) has its unique magnetic fingerprint
- A magnetic read-head detects the position

Advantages

- Very high resolutions possible ($0.7 \cdot 10^{-4}$ deg)
- Enables absolute position measurement on **hollow shafts**

Disadvantages

- Complex and expensive
- Very high demands on distance and centricity



Magnetic off-axis absolute encoders in different sizes as used in ARMAR-6

Potentiometer (Variable Resistor)

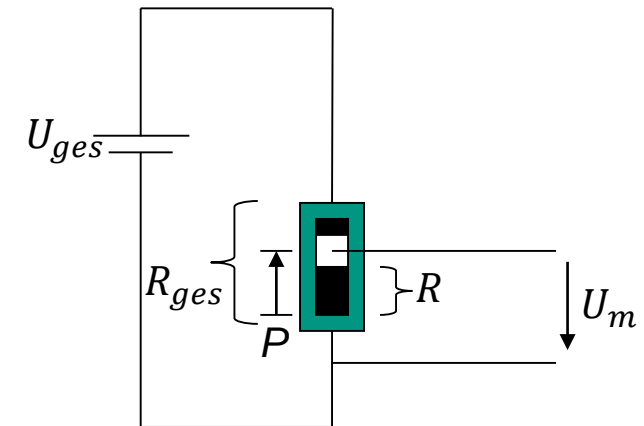
- Resistors that change their resistance according to the position of a slider
- Implemented as variable voltage divider with three connections
- The voltage ratio can be converted into a position
- Well suited for low-cost applications with low demands on compactness and precision



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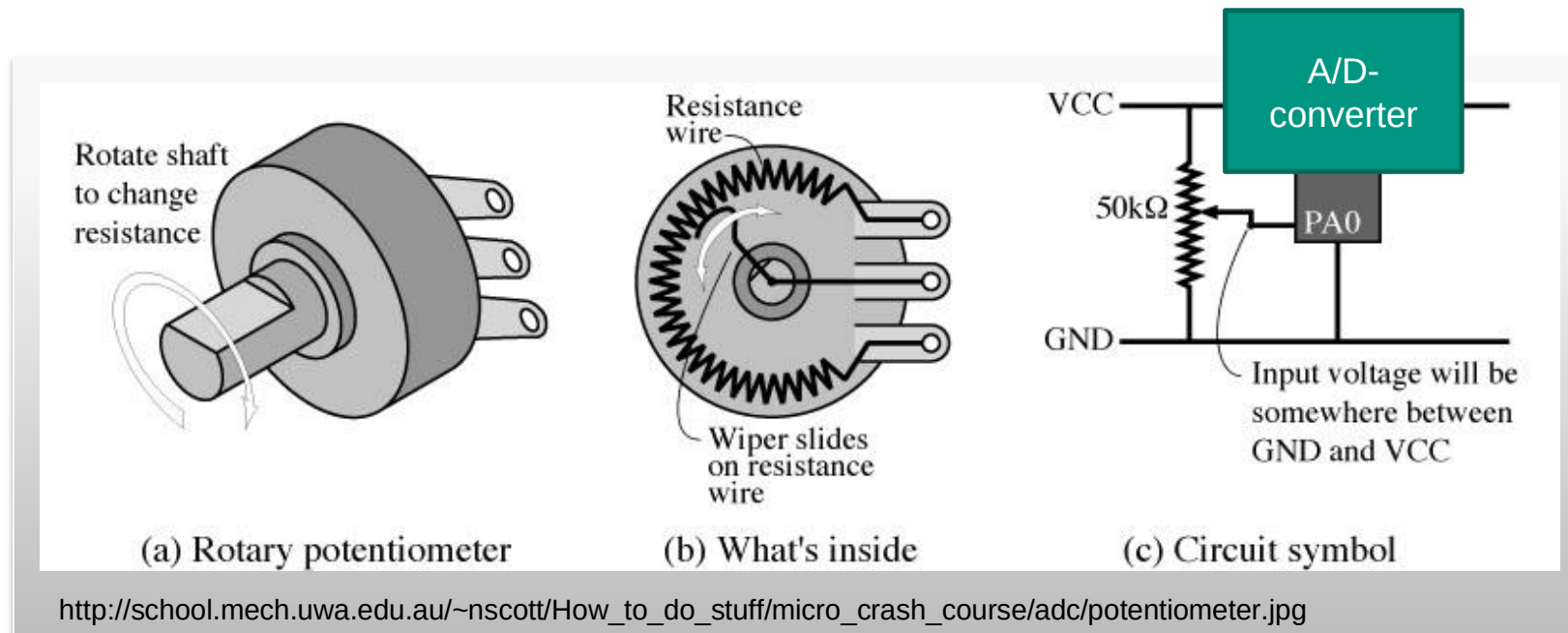
Deriving the position from the voltage U_m

- Across the entire potentiometer drops a voltage of U_{ges}
- Voltage U_m scales proportionally to P from $0V$ to U_{ges}
- $$U_m = \frac{R}{R_{ges}} U_{ges} \sim P$$



Potentiometer – Connections

- Potentiometer are analog sensors
- The voltage signal requires digitization
- The resolution of the analog-to-digital converter (ADC) determines the resolution of the sensor system



Potentiometer – Styles

Rotary potentiometer

- Absolute position measurement of rotational motion
- Sub-styles:
 - Single-Turn: Range of maximum one revolution (most common)
 - Multi-Turn: Range of up to 10 (or even more) revolutions

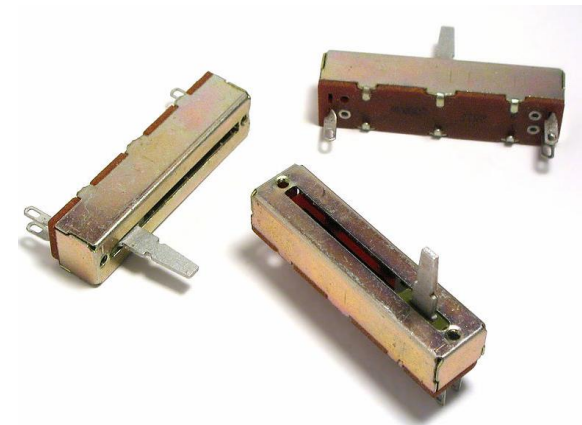
Linear potentiometer

- Absolute linear position measurement
- Available in a large variety of measurement ranges



https://partnership.bourns.com/bu/bu_prec.shtml

Multi-turn rotary potentiometer with a measurement range of ten revolutions



<https://de.wikipedia.org/wiki/Potentiometer#/media/File:Faders.jpg>

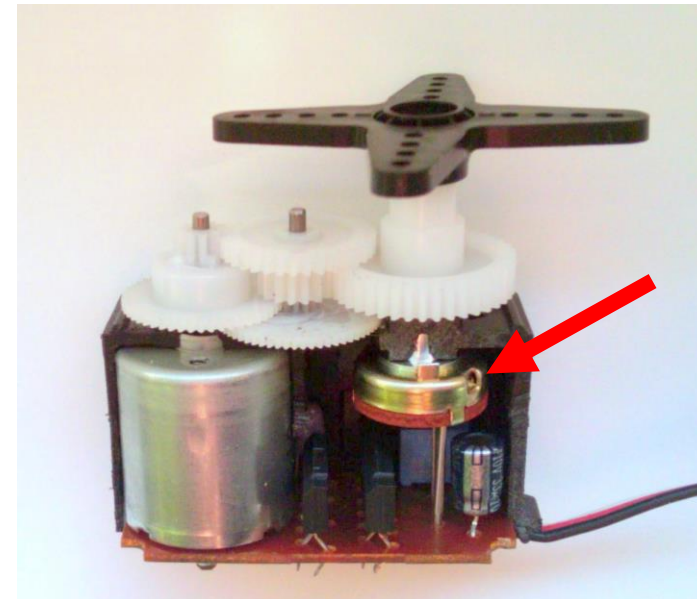
Potentiometer – Applications

- Often used as absolute position sensors in low-cost applications
- **RC (radio control) servo motors**
- User interfaces (rotary dials)



© RobotShop inc. (2020)

Hexapod robot equipped with RC servo motors



Cross-section of a RC servo motor with potentiometer for position feedback at the output shaft

Draw Wire Sensors (Absolute Encoders)

- Suitable **for long linear movements** (e.g. hydraulic pistons)
- Linear motion unwinds a string from a rotating drum
- A potentiometer, incremental or absolute encoder measures the rotation of the drum

Advantages

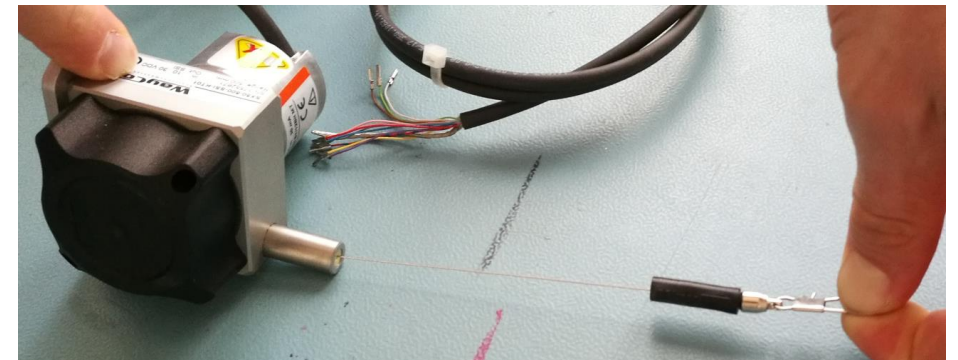
- Precise measurement of long linear displacements
- Range: 50mm – 5000 mm
- Easy to integrate
- Robust

Disadvantages

- Relatively large
- Actuator must compensate for the restoring force of the drum spring



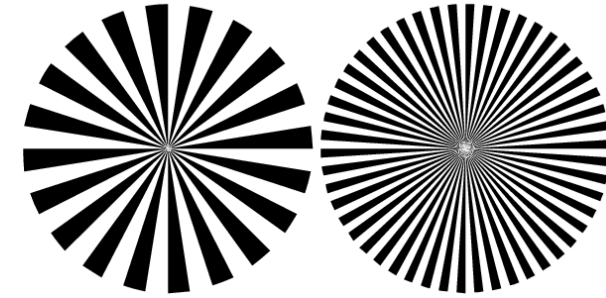
© Micro-Epsilon (2020)



Draw wire sensor used to measure the torso extension of ARMAR-6

Summary: Internal Position Sensors

- Internal position sensors sense the joint displacements of the robot:
 - **Rotary joints** → **Angular sensors**
 - **Prismatic joints** → **Distance sensors**
- Three most widely spread sensor technologies for rotary position encoders are:
 - Optical
 - Magnetic
 - Potentiometers
- One can differentiate between different rotary encoders by the way they are mounted:
 - On-axis
 - Off-axis



Optical encoder discs



Magnetic off-axis position sensor

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Interfaces for Absolute Encoders

Interfaces for Digital Absolute Encoders – I2C

- Inter-Integrated Circuit (I²C)
- Synonym: TWI (Two Wire Interface)
- Developed 1982 by Philips Semiconductors

Operating principle

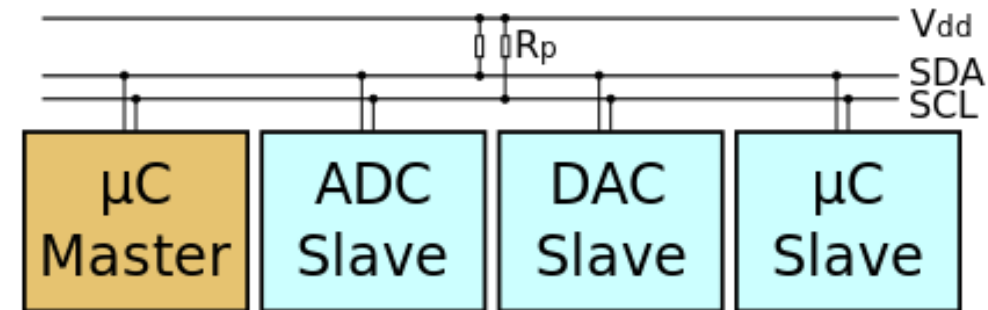
- **Two-wire serial communication protocol** with two physical lines: serial clock (SCL) and serial data (SDA)
- Master-Slave Bus for communication between ICs
- Each slave has unique address (**7-10 bit address space**)
- Bidirectional

Advantages

- Simple wiring even for large number of devices/sensors

Disadvantage

- Comparatively slow (standard: 100 kbit/s)
- In “Ultra Fast-mode”: 5 Mbit/s (Unidirectional)



<https://de.wikipedia.org/wiki/I%C2%B2C>

I²C-network with one master and three slaves

Interfaces for Digital Absolute Encoders – SPI

- Serial Peripheral Interface (SPI)
- Developed 1980 by Motorola
- Relatively loose definition with lots of possible variations

Operating principle

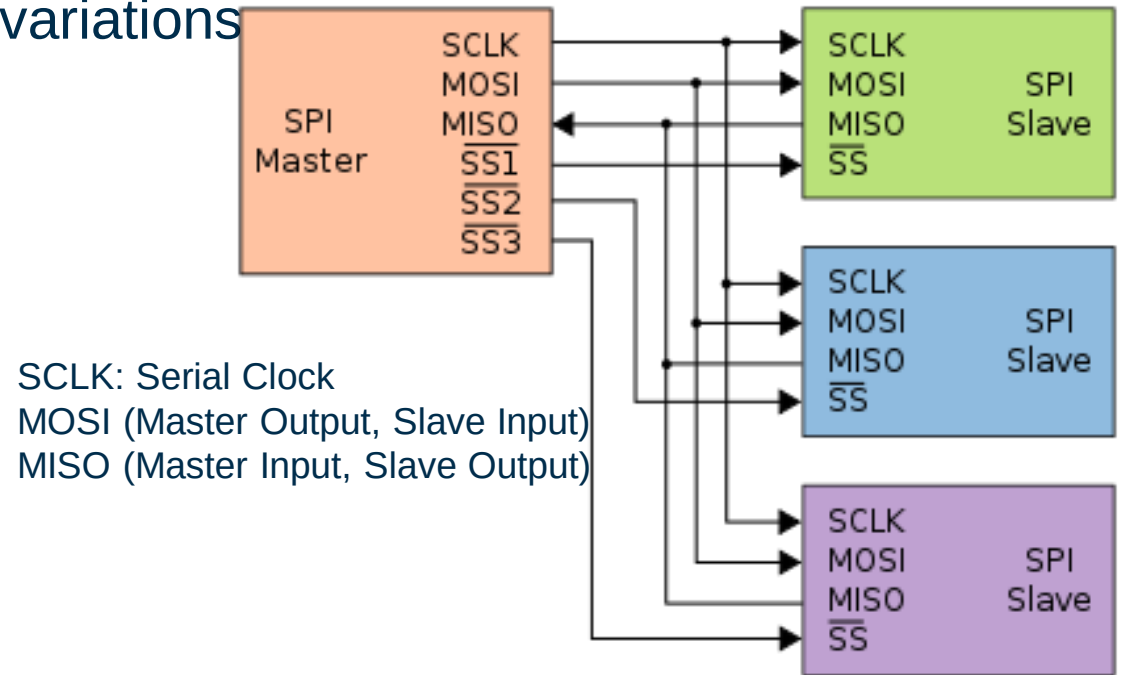
- **Synchronous serial communication protocol**
- Master-Slave bus with **four physical lines**
- Each slave is addressed by the master via a dedicated *chips select signal* (Slave Select: SS)

Advantages

- Very simple on the software side (because arbitration via chip select signals)
- Speed defined by the master clock

Disadvantage

- Higher circuit complexity compared to I²C



https://de.wikipedia.org/wiki/Serial_Peripheral_Interface

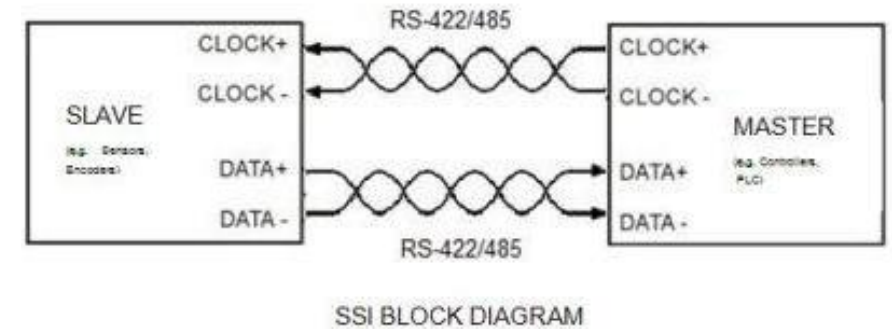
SPI-network with master and three slaves
(star topology)

Interfaces for Digital Absolute Encoders – SSI

- „Synchron Serielle Interface“ (SSI)
- Developed in 1984 by Max Stegmann GmbH

Operation principle

- **Synchronous, point-to-point, serial communication**
- Uni-directional point-to-point connection
- **Two twisted wire pairs** (data and clock)
- Twisted wire pairs for differential signal transmission, i.e. transmitting information using two complementary signals



https://de.wikipedia.org/wiki/Synchron-Serielle_Schnittstelle

Advantages

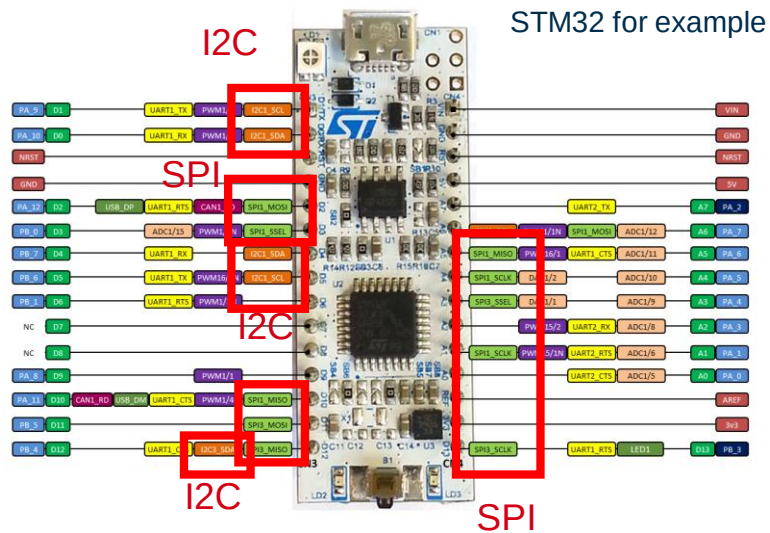
- High electrical robustness (designed for reliability in industrial applications)
- Very simple transmission protocol

Disadvantages

- High cabling complexity due to twisted pairs
- Limited topologies (point-to-point)

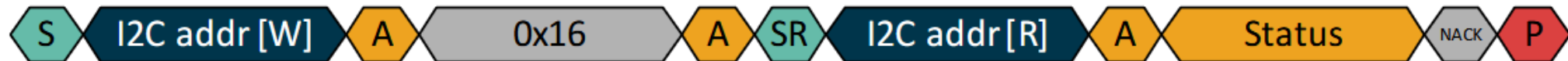
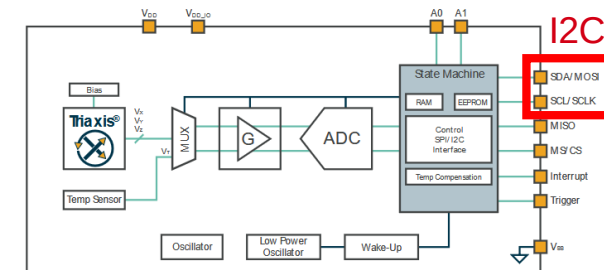
Interfaces for Digital Absolute Encoders – Implementation

- How to read out the data of a sensor?
 - Almost all microcontrollers provide multiple I2C and SPI interfaces (e.g. STM32)

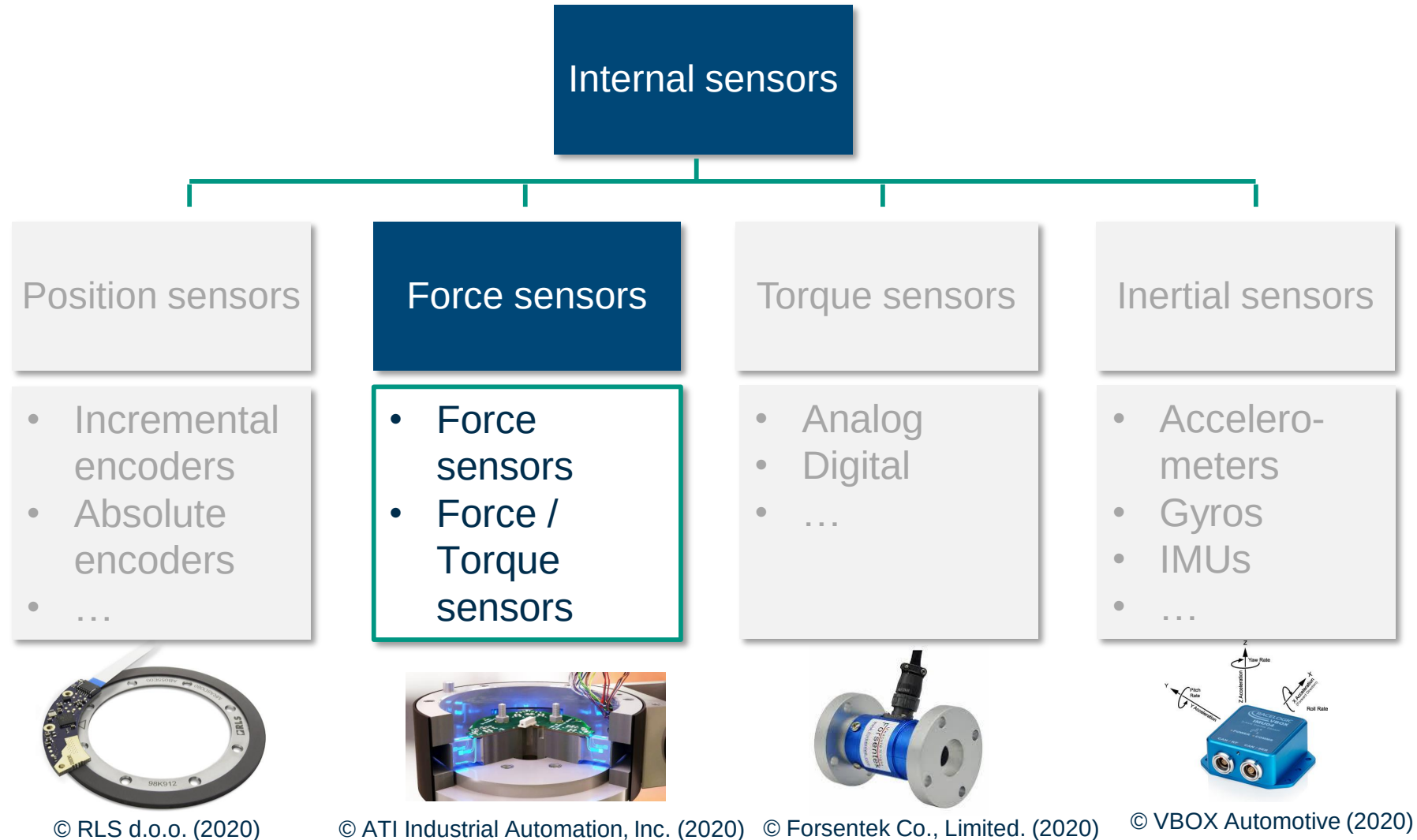


- Connect sensor (2 data-lines for I2C, don't forget pullup resistors...)
- Low level FW-driver (provided by ST) for sending and receiving data:
`HAL_I2C_Master_Transmit(&hi2c, ...);`
`HAL_I2C_Slave_Receive(&hi2c, (uint8_t *)RX_Buffer, ...);`
- Datasheet of the encoder “explains” the commands, data-format, etc.

- Read command for the Melexis MLX90393 (Absolute Position Sensor IC)
(<https://media.melexis.com/-/media/files/documents/datasheets/mlx90393-datasheet-melexis.pdf>)



Internal Sensors – Overview



Force Sensors

- Force sensors allow to measure forces
 - in the body of the robot (internal)
 - between the robot and the environment

Types

- 1D force sensors
- 3D force sensors
- 6D force/torque sensors
- ...

Applications

- Measurement of internal forces (e.g. in Bowden cables)
- Measurement of contact forces (e.g. ground contact, open/close doors, haptics, tactile-servoing, ...)
- Measurement of interaction forces (Human robot interaction/collaboration...)



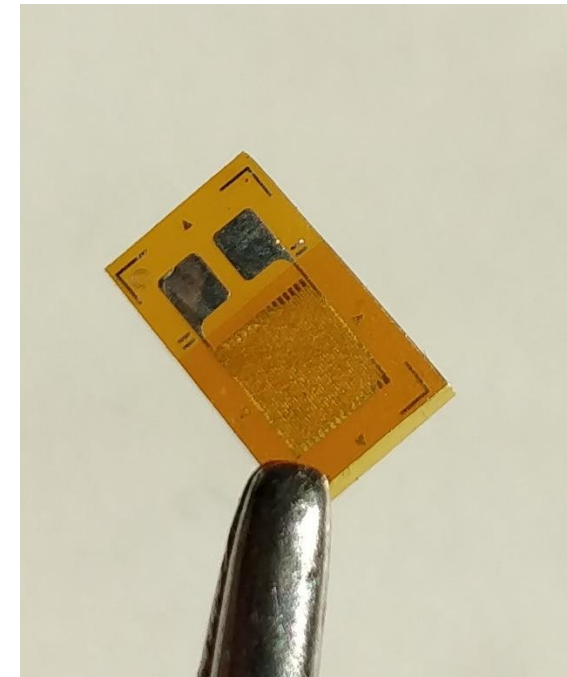
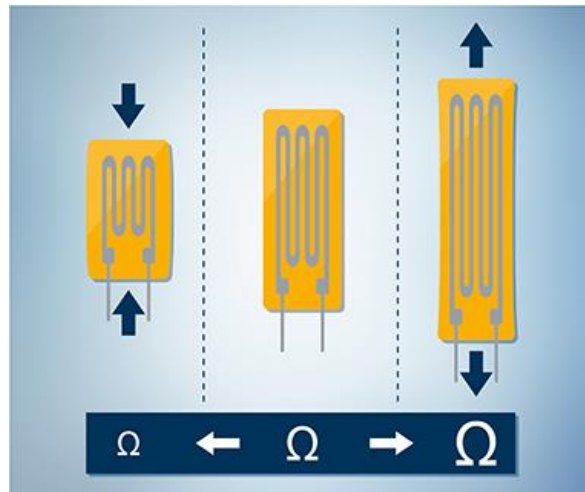
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Preliminaries

Strain gauges, AD conversion, measuring resistance

Resistive Strain Gauges

- Strain gauges transform micro-deformations (due to strain) into a change of their electrical resistance: **Piezoresistive Effect**
- Effect can be amplified by a series of connection of several windings
- Using a **measurement bridge**, the change in resistance is converted to a change in voltage
- This voltage can be amplified and measured



Piezoresistive Effect (I)

- The underlying principle of strain gauges is the fact that materials change their electrical resistance when deformed
- Causes for this:
 - Change in **geometry** (occurs for all materials; effect is small)
 - Change in the **specific resistivity** (effect is large effect in piezo-electric materials)

Piezoresistive Effect (II)

- Electric resistance of the unloaded strain gauge depends on the specific resistivity ρ , the length l and the cross-section A (with D being the wire diameter):

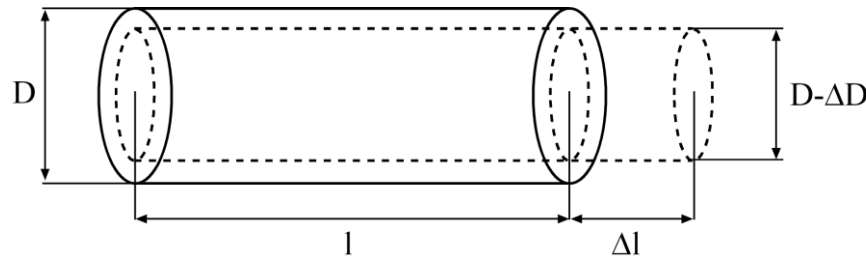
$$R = \rho \cdot \frac{l}{A} = \rho \cdot \frac{4l}{D^2\pi}$$

- Under strain load, **length**, **cross-section**, and **specific resistivity** of the wires change:

$$R + \Delta R = (\rho + \Delta\rho) \cdot \frac{4(l + \Delta l)}{(D - \Delta D)^2\pi}$$

- Re-arranging the equation above and introducing the elongation ϵ as well as the sensitivity coefficient k , one obtains the relation between elongation and change in resistance:

$$\frac{\Delta R}{R} = k \cdot \frac{\Delta L}{L} = k \cdot \epsilon$$



Measuring Resistance

An electrical resistance can be measured as a voltage using a **voltage divider circuit**.

Example

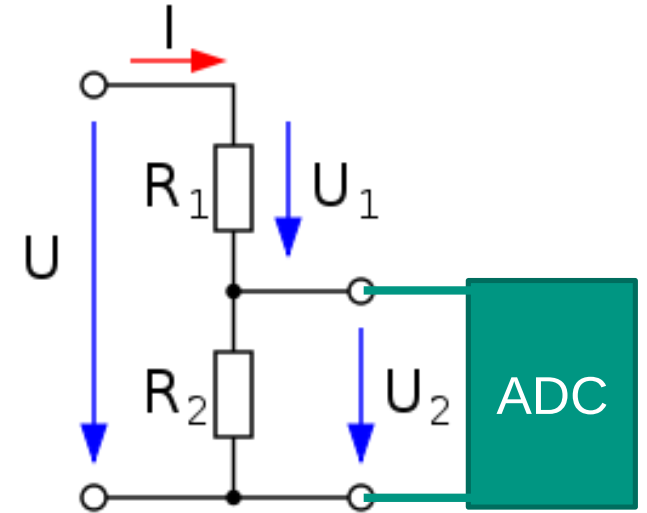
- The variable resistor R_2 shall be determined by measuring the voltage U_2
- R_1 is a fixed reference resistor

$$\frac{U_2}{U} = \frac{R_2}{R_1 + R_2}$$

Problem

- The ADC measurement range is U
- If the **change in resistance of U_2 is small compared to U** only a small fraction of the ADC's range will be exploited

→ **lower resolution**, **Solution:** Wheatstone bridge



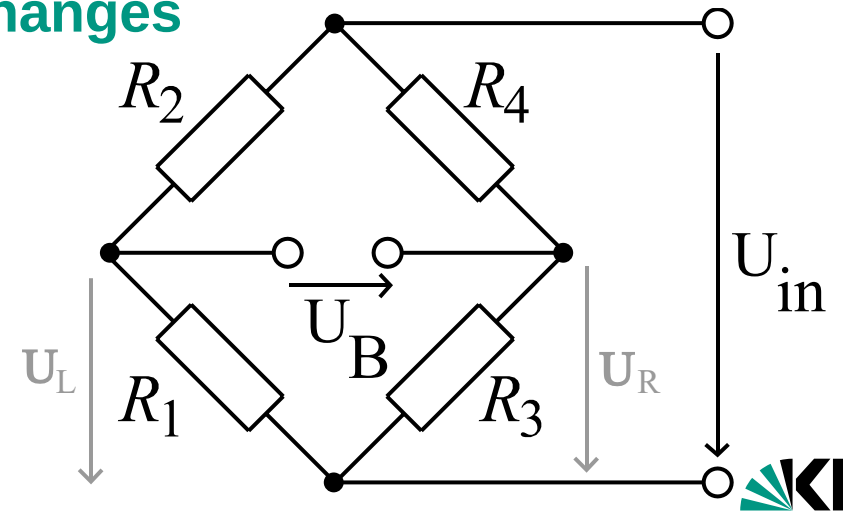
<https://de.wikipedia.org/wiki/Spannungsteiler#/media/Datei:Einfacher-unbelasteter-Spannungsteiler.svg>

Wheatstone Bridge

- A Wheatstone bridge in conjunction with an amplifier allows to measure **very small changes in resistance with high resolution**
- It consists of two voltage dividers (R_1 and R_2 as well as R_3 and R_4) that operate from the same supply voltage (U_{in})
- An ADC measures the **differential voltage U_B** („detuning“) of the two voltage dividers (not with reference to GND)
- $U_B = U_L - U_R = U_{in} \left(\frac{R_1}{R_1 + R_2} - \frac{R_3}{R_3 + R_4} \right)$
- This small voltage difference can be amplified (AMP) to cover the entire range of the ADC
→ **Using the entire range of the ADC even for small changes**

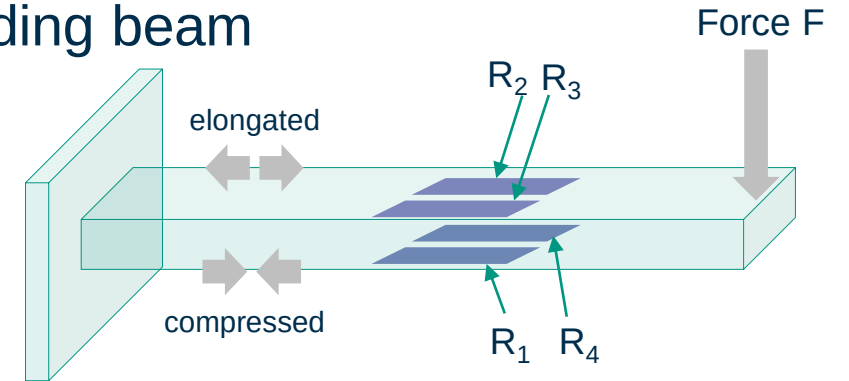
Smart arrangement of the resistors/strain gauges allows **disturbance suppression/compensation**

- Temperature compensation
- Torsion deflection compensation
- Elongation compensation



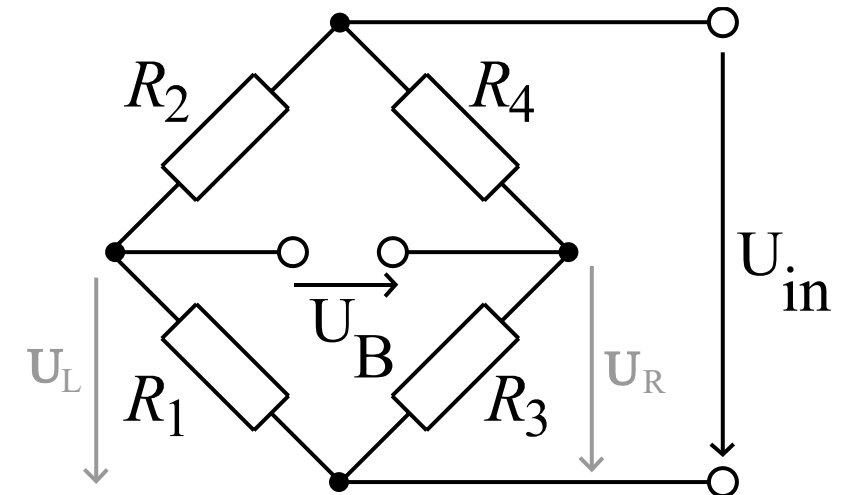
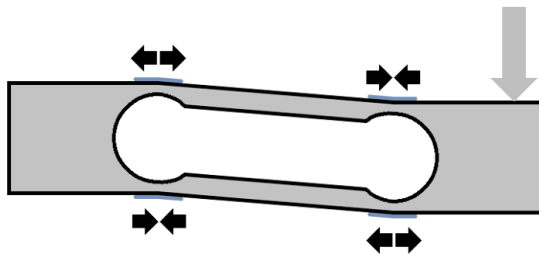
Wheatstone Bridge: Force Sensor

- 4 strain gauges used for measuring force F at a bending beam
- For positive force F :
 - R_2, R_3 getting larger (elongated) $\rightarrow R + \Delta R$
 - R_1, R_4 getting smaller (compressed) $\rightarrow R - \Delta R$
- Under load: $U_B = U_{in} \left(\frac{R_1}{R_1 + R_2} - \frac{R_3}{R_3 + R_4} \right) =$



$$U_{in} \left(\frac{R - \Delta R}{R - \Delta R + R + \Delta R} - \frac{R + \Delta R}{R + \Delta R + R - \Delta R} \right) = U_{in} \cdot \frac{\Delta R}{R}$$

- Double bending beam: Higher sensitivity

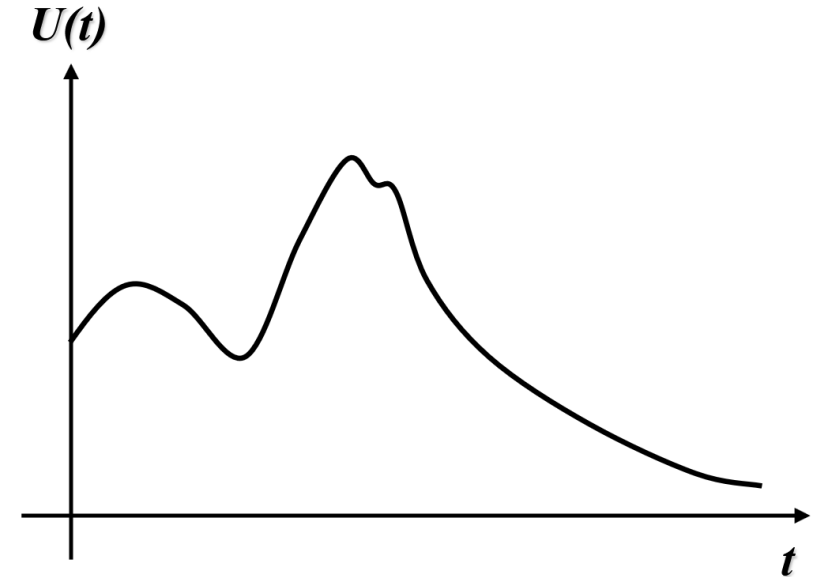


Analog-to-Digital Conversion

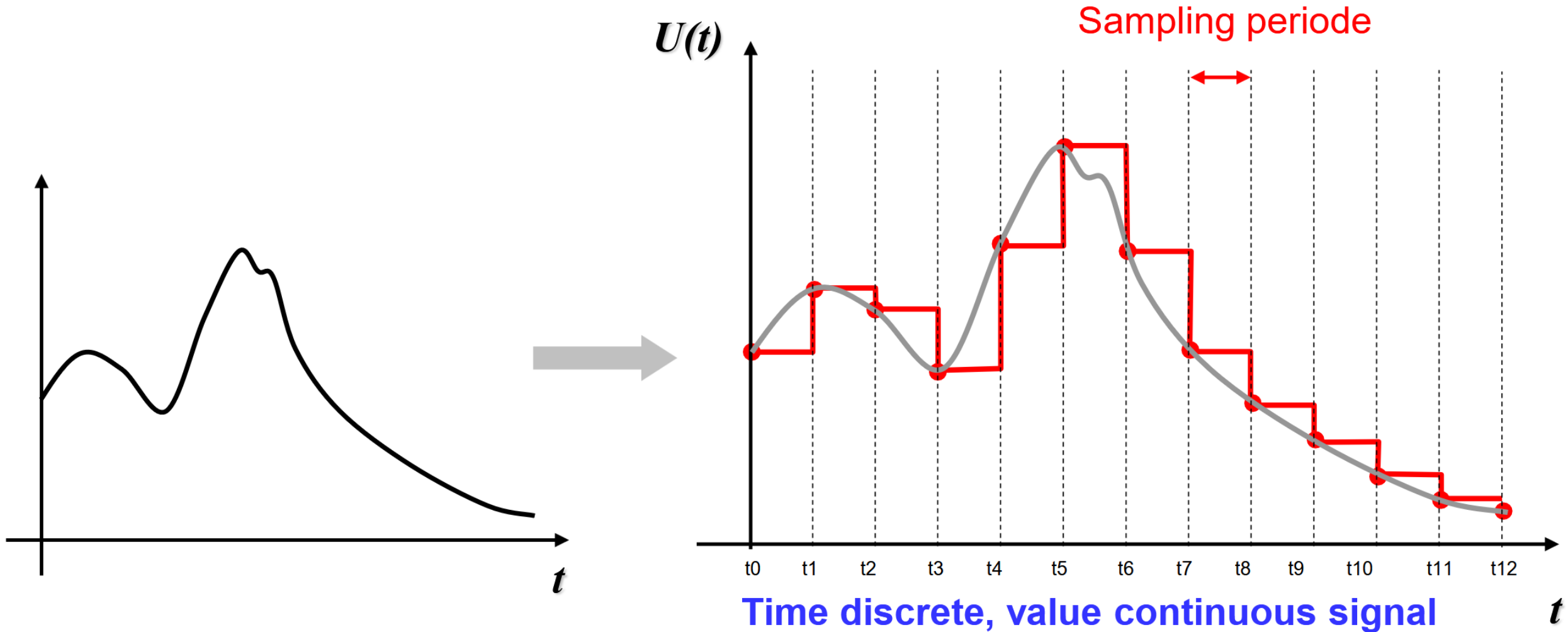
- For **conversion** from continuous signal $\rightarrow$ Digital signal:

Nyquist–Shannon Sampling Theorem

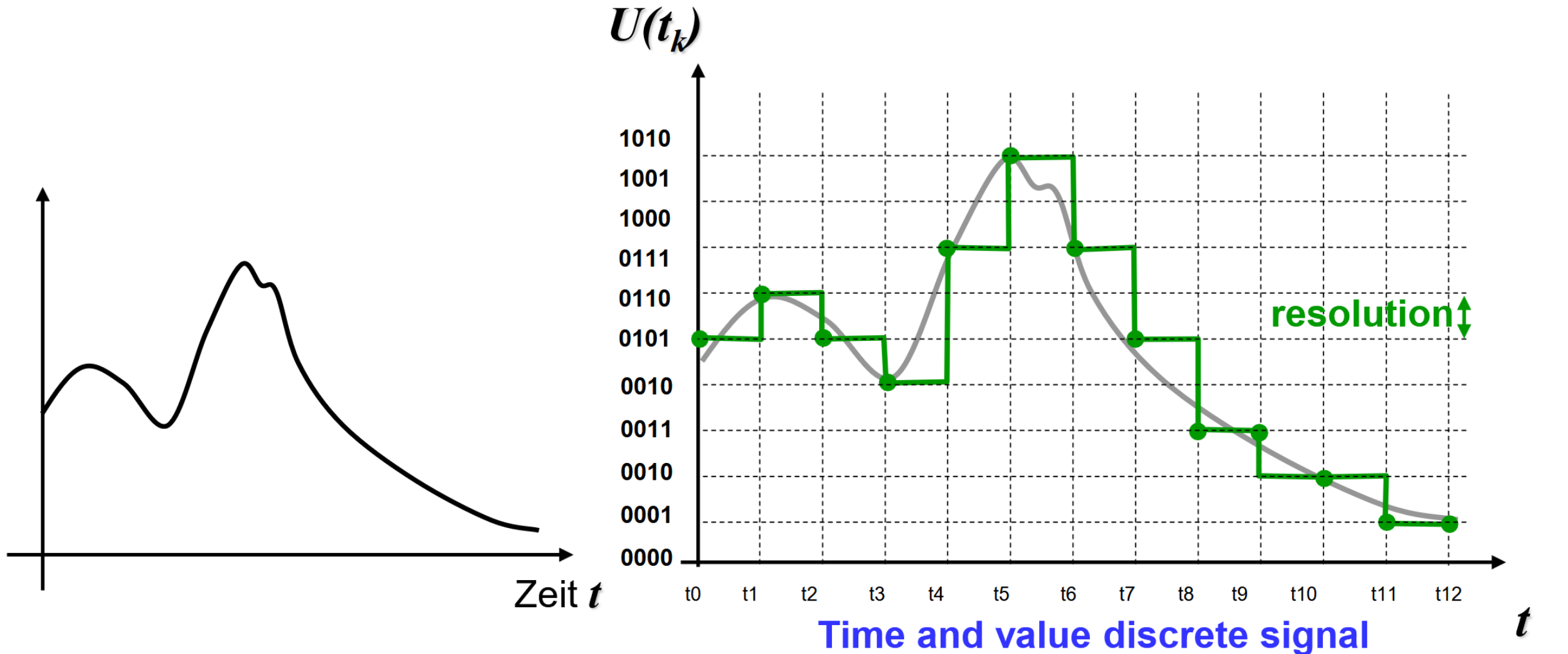
Reproduction of the original signal is only possible if the sampling rate is higher than twice the highest frequency of the signal.



Analog-to-Digital Conversion



Analog-to-Digital Conversion

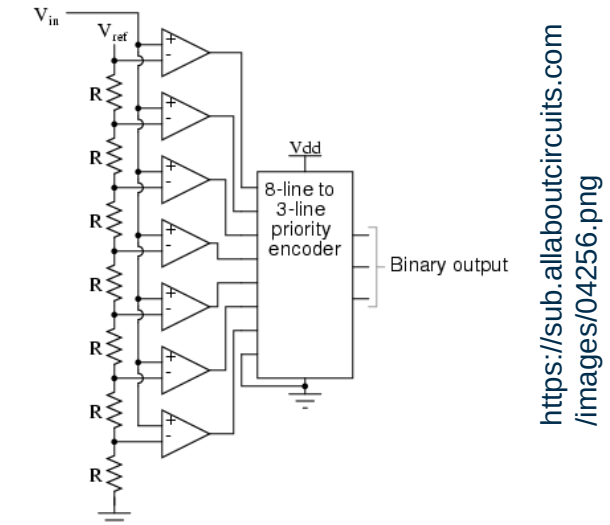


Analog-to-Digital Conversion (ADC)

- Crucial component in many sensor systems
- Digitization of a voltage V_{in} in relation to a reference voltage V_{ref}
- **Characteristics** (among others):
 - Resolution (in Bits)
 - Signal-to-Noise
- **Example ADS1220** (low-noise differential ADC used for torque-sensing in ARMAR-6)
 - 24-Bit resolution
 - Integrated analog and digital signal processing
 - Integrated temperature sensor for temperature compensation
 - SPI-Interface

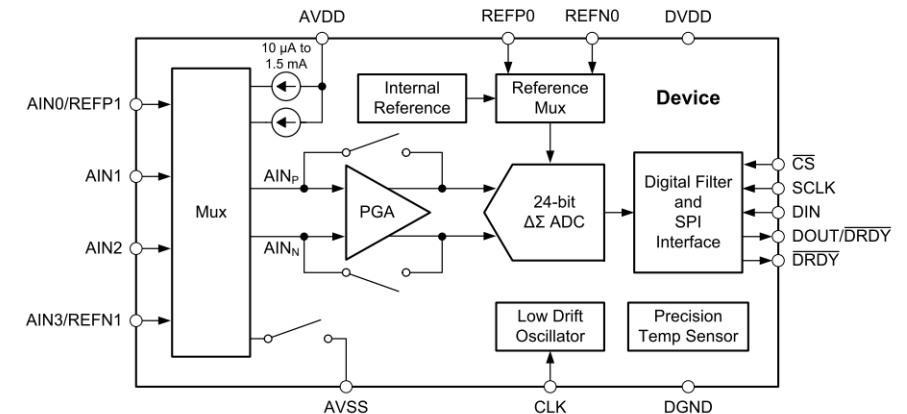


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<https://sub.allaboutcircuits.com/images/04256.png>

Operation principle of an 8-Bit ADC



Block Diagram of ADS1220 Chip Internals

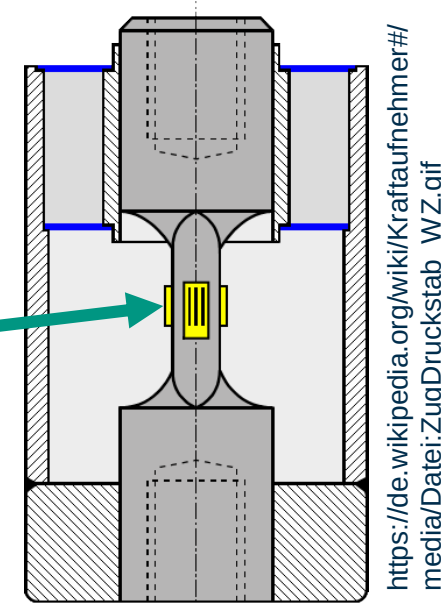
© Texas Instruments Incorporated (2020)

1D Force Sensors

1D Force Sensors

- **Principle of operation:** A sensor element deforms when subjected to mechanical load
- The deformation is converted to an electrical signal either
 - **Resistively (with strain gauges)**
 - Capacitively (Microelectromechanical systems: MEMS)
- The force can be calculated from the electrical signal
- Within the specified range, the produced signal is proportional to the load

The sensors described in this lecture are analog and require subsequent A/D conversion of the sensors signal



1D force sensor (working principle and example)

Force Sensing Resistors (FSR)

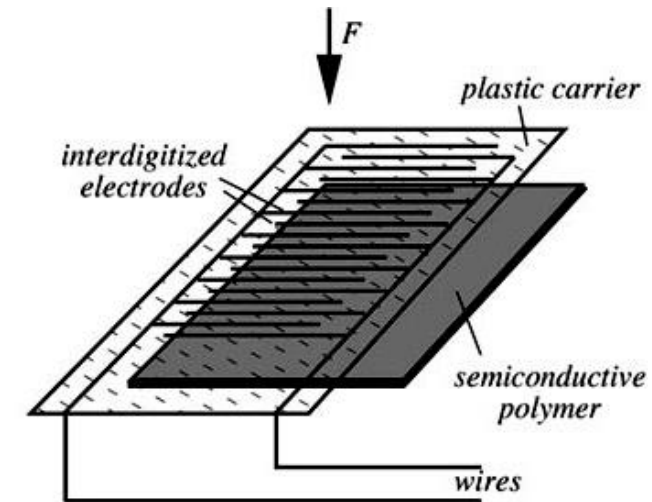
- FSR are electrical resistors that change their **resistance** due to mechanical deformation
- Large change in resistance, used in voltage dividers

Principle of operation

- The two connectors lead to two interwoven „combs“ that are not connected
- The combs are covered by a conductive polymer
- When pressed, the conductive polymer shorts the two traces together with a resistance that depends on applied force



<https://www.antratek.de/force-sensing-resistor-0-5>



From Fraden, J.: Handbook of Modern Sensors

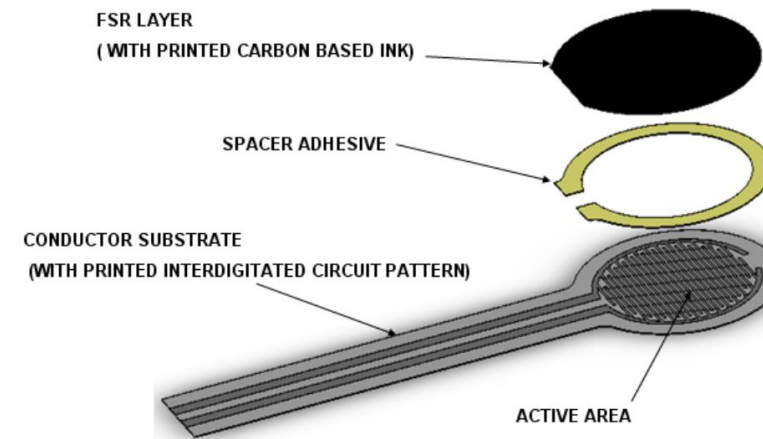
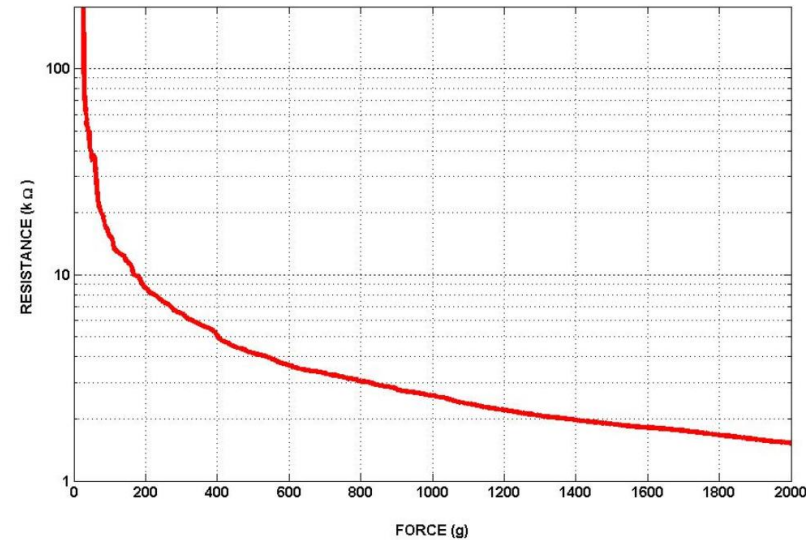
Force Sensing Resistors (FSR)

Advantages

- Change in resistance is larger than in strain gauges
 - **No need for differential measurement**
 - Can be interfaced via a voltage divider and a simple microcontroller with integrated ADC
- **Very low cost**

Disadvantages

- Very low accuracy (rather qualitative than quantitative measurement)
- Measured value depends on many factors, not only on the load



https://www.rapidonline.com/pdf/182546_in_en_01.pdf

6D Force/Torque Sensors

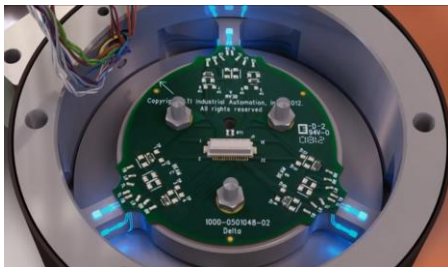
6D Force/Torque Sensors

- Measure the **3D force vector** and the **3D torque vector**
- Commonly used in **robotic end-effectors** (hands, feet)
- Produce **six analog signals** (integrated Wheatstone bridges) that need to be digitized and converted into force and torque values

$$(f_x, f_y, f_z, \tau_x, \tau_y, \tau_z)^T = \mathbf{M} \cdot (analog\ signals)^T$$

With $\mathbf{M}$ being the 6x6 calibration matrix

- **Very Expensive** (ca. 5000€)

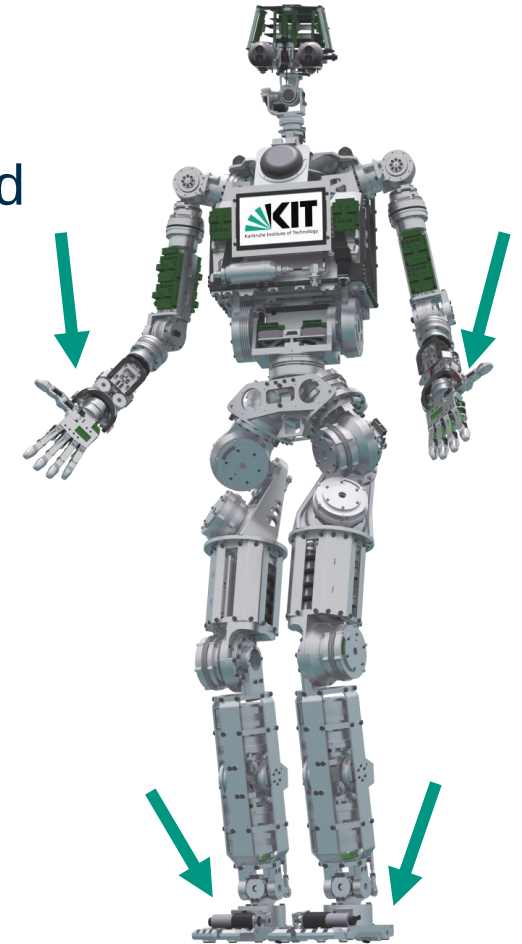


© ATI Industrial Automation, Inc. (2020)

Animated inner view of a force/torque sensors with highlighted strain gauges



© ATI Industrial Automation, Inc. (2020)

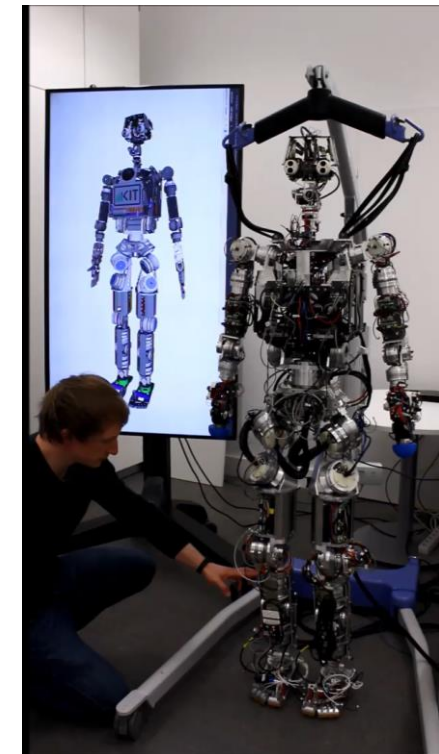


6D Force/Torque Sensors – Application

- Precise measurement of the **ground reaction wrench** (forces and torques)
 - Enables computing the pushing forces along the robot's body
- Contact force/torque of the hand with the environment
 - Enables compliance adaptation

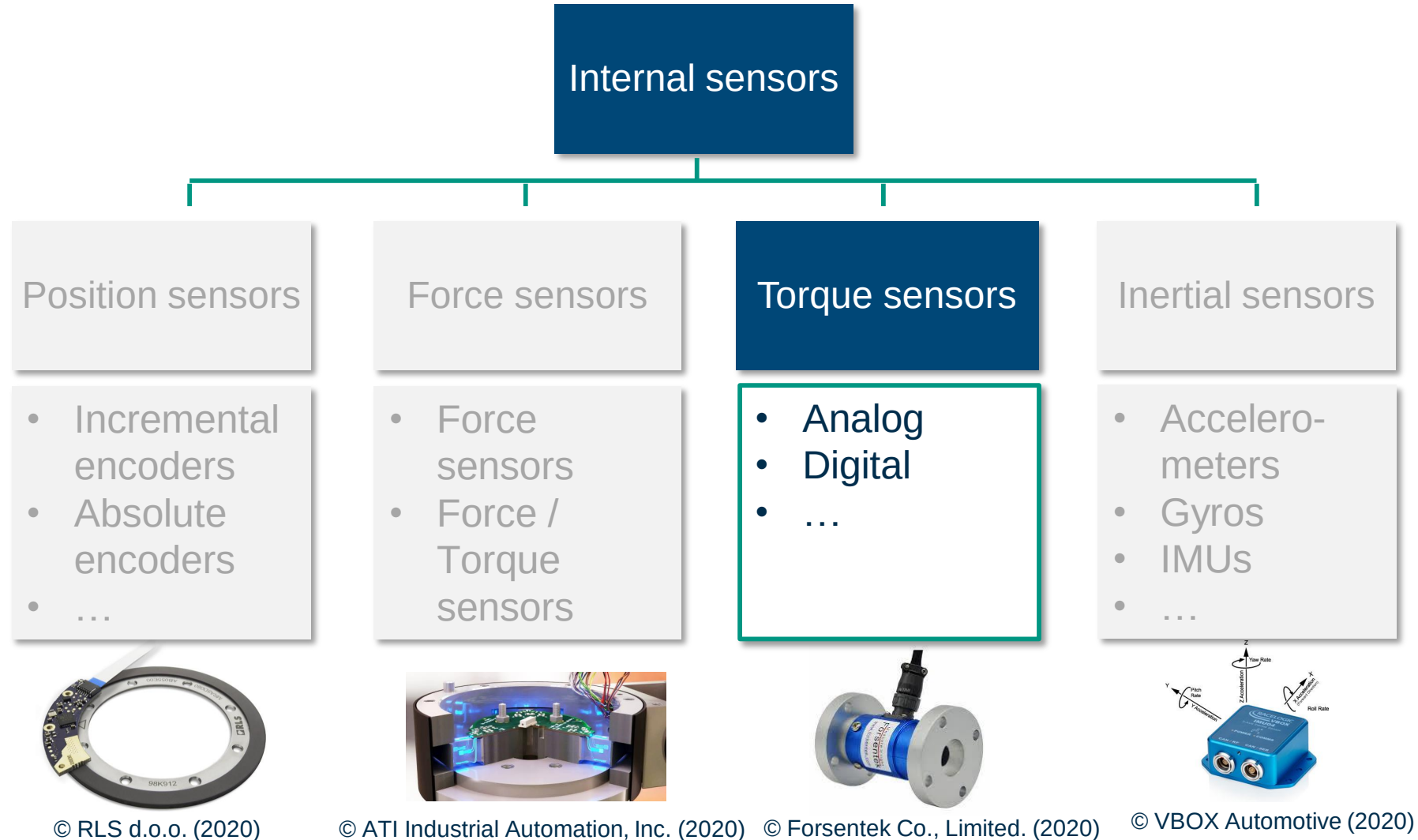


In the wrist: for compliance adaptation



In the ankle: reaction forces

Internal Sensors – Overview



Torque Sensors

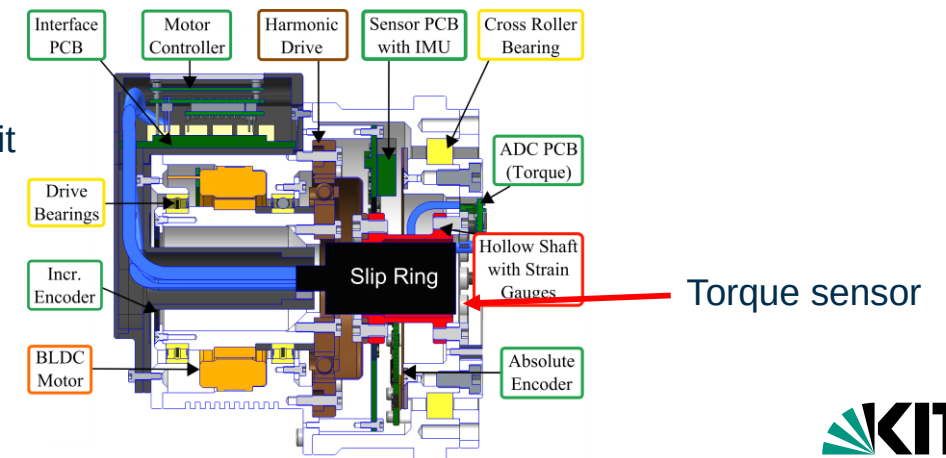
- Measure the torques in the robot joints
- Necessary as feedback sensors in **joint torque control**
 - For human-robot interaction
 - For compliant reaction to external contacts

Operation principle

- Determine the **torsional deformation** of the output shaft (between the gear and the output flange)
 - **strain gauges**
 - high precision position encoders



ARMAR-6 sensor-actuator-controller-unit



Torque Sensors – Analog I

Spoke wheel type

- Sensors consists of a milled spoke wheel
- The spokes act as **bending beams**
- Bending of the spokes is proportional to the torque acting **between the inner and the outer ring**
- **Strain gauges** on the spokes measure the bending
- The spokes are wired to form a complete **Wheatstone bridge**
- Digitization of the analog signal with **internal or external ADC**



Spoke wheel type torque sensor of the DLR light-weight arm

Torque Sensors – Analog II

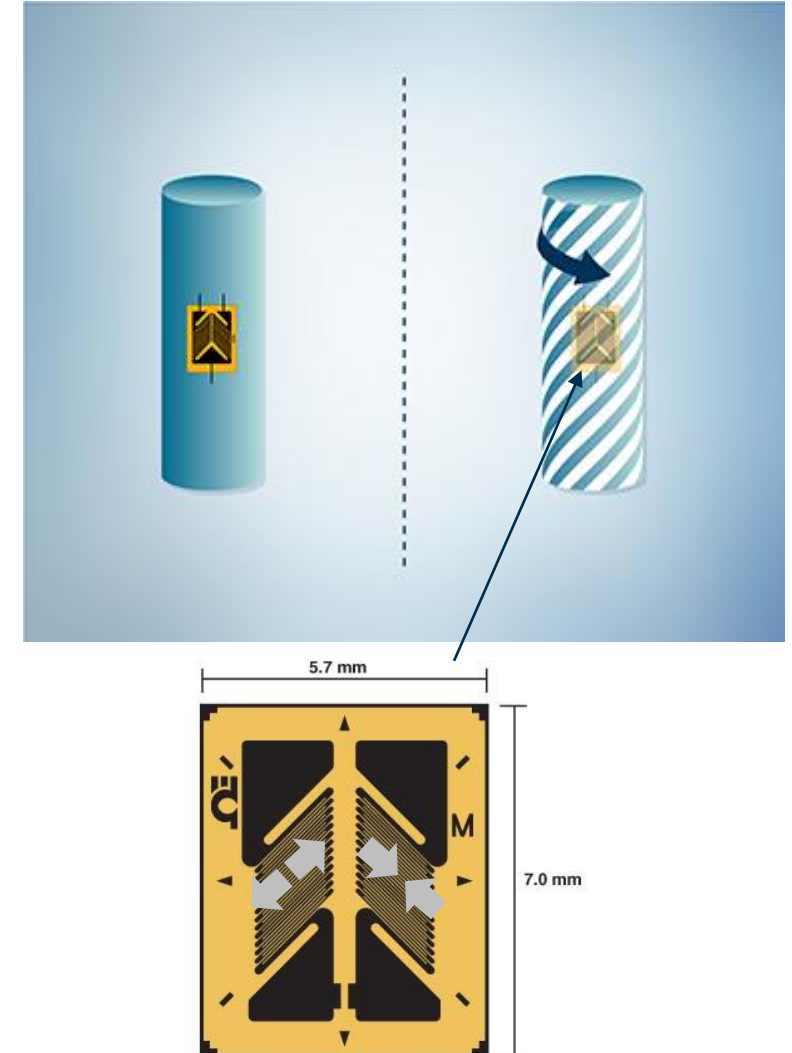
Torsional shaft type

- The sensor consist of a **thin-walled hollow shaft** between the gear and the output flange
- The hollow shaft undergoes **torsional deformation** due to the acting torque
- **Strain gauges** detect the deformation (while compensating temperature and bending effects) and are wired as **Wheatstone bridge**
- The voltage is digitized using a **differential ADC** and linearly converted into a torque value

Advantages

- Small installation space
- High precision
- High stiffness

<https://www.hbm.com/10716/how-does-a-torque-transducer-actually-work/>



Torque Sensors – Digital

Torsional shaft type

- Torsional shaft at the actuator output (like in the analog case)
- Deformation is measured using a digital **absolute encoder** between the two flanges of the hollow shaft
 - The encoder is mounted on the output side
 - The magnetic disc is mounted on the input side

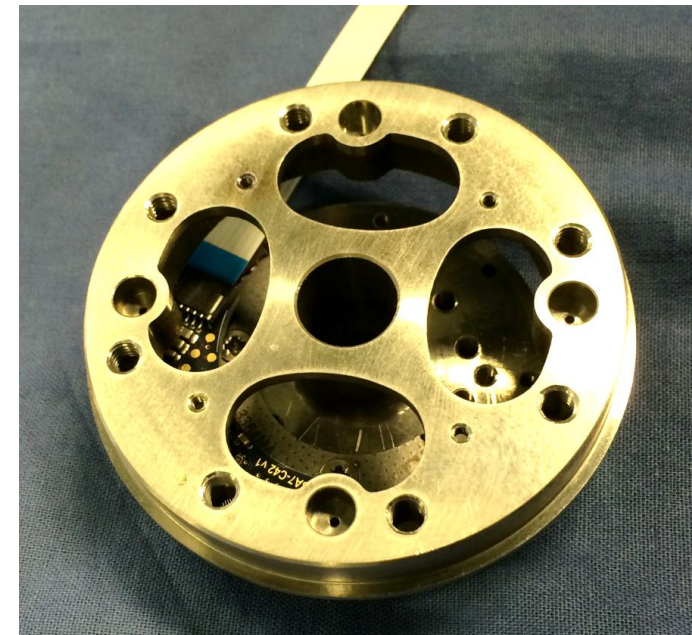
Advantage

- No need for A/D conversion

Disadvantage

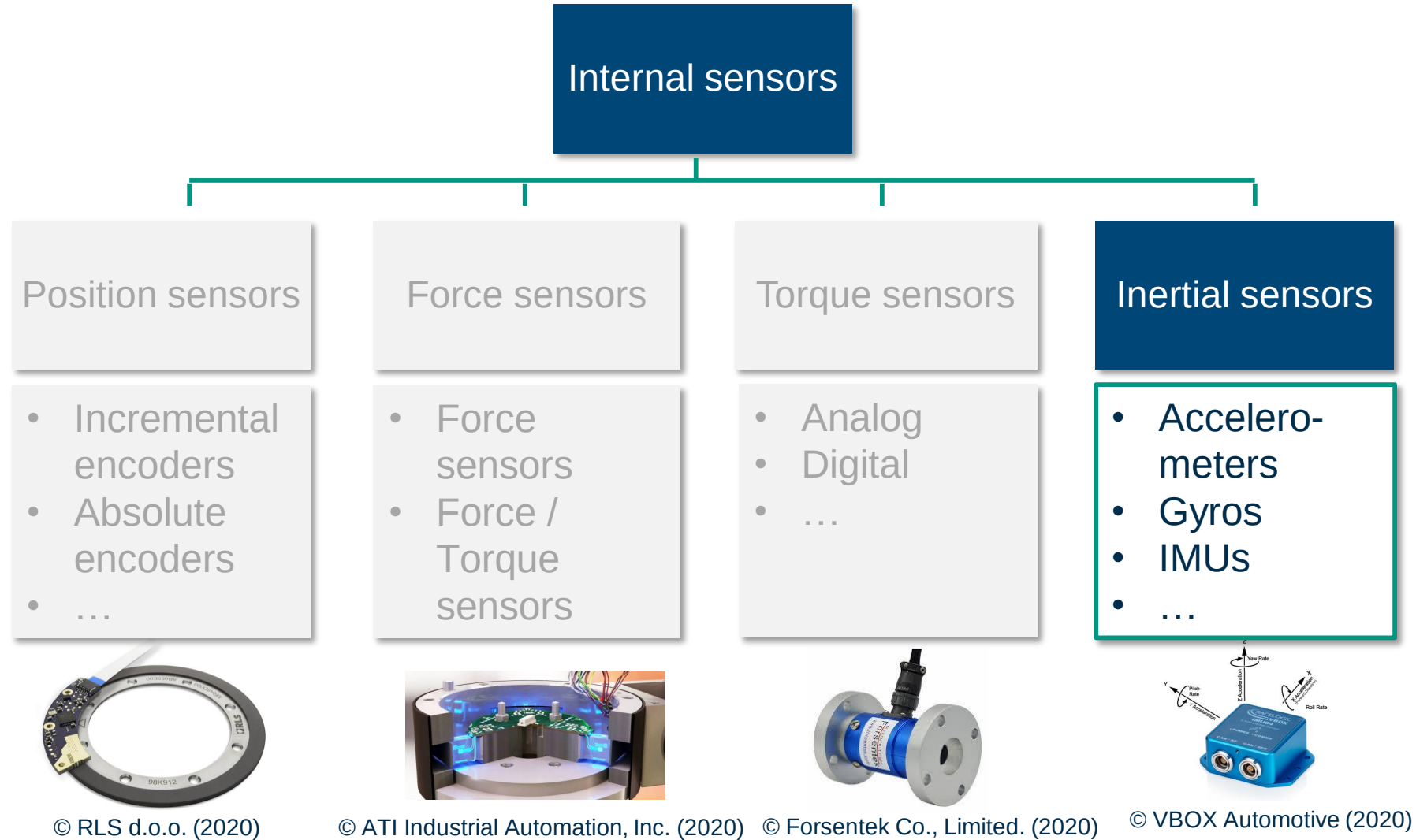
- Resolution limited by encoder resolution (not utilizing the encoder's full range)

Image taken from: Baccelliere, Lorenzo, et al. "Development of a human size and strength compliant bi-manual platform for realistic heavy manipulation tasks." Intelligent Robots and Systems (IROS), 2017 IEEE/RSJ International Conference on. IEEE, 2017.



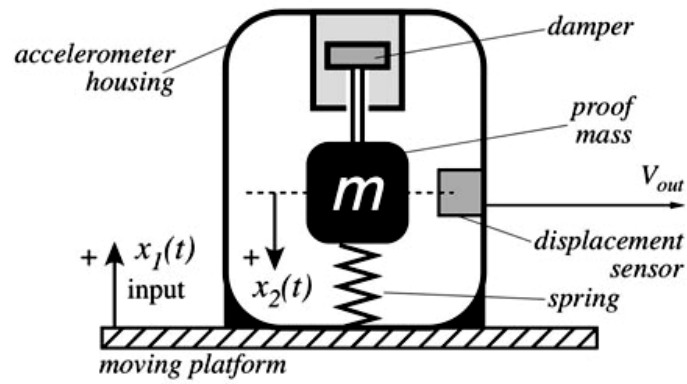
Encoder-based torque sensor

Internal Sensors – Overview

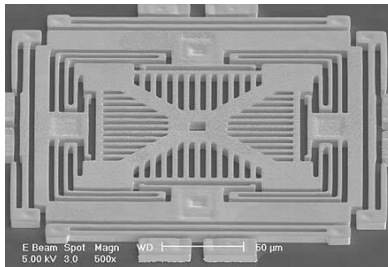


Inertial Sensors

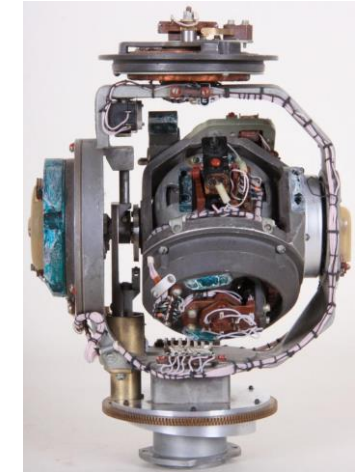
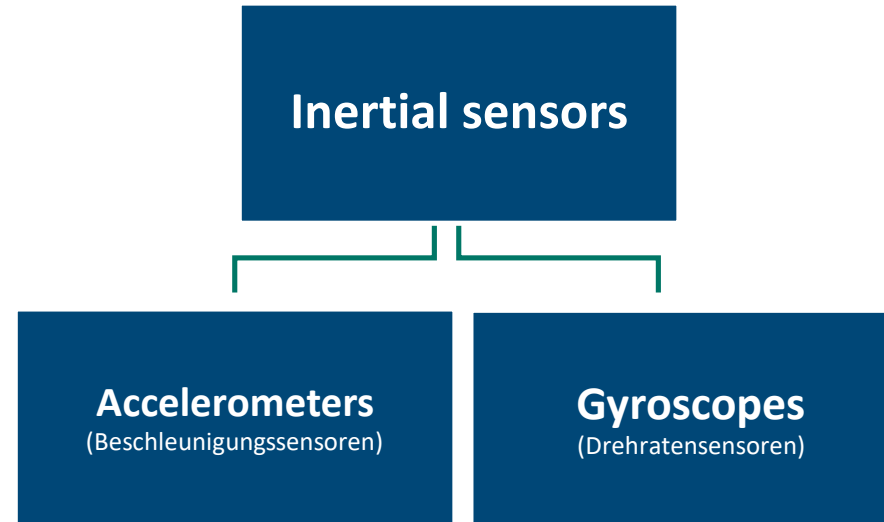
Sensors that measure effects based on the inertia of a proof mass



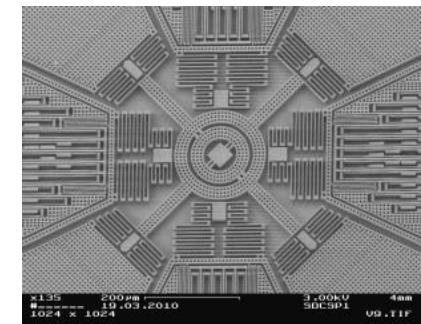
From Fraden, J.: Handbook of Modern Sensors



<http://secondelmb.free.fr/edc2/activites/act3.html>



https://www.ostron.de/out/pictures/z3/kreiselbaugruppe_mig21_03_z3.jpg

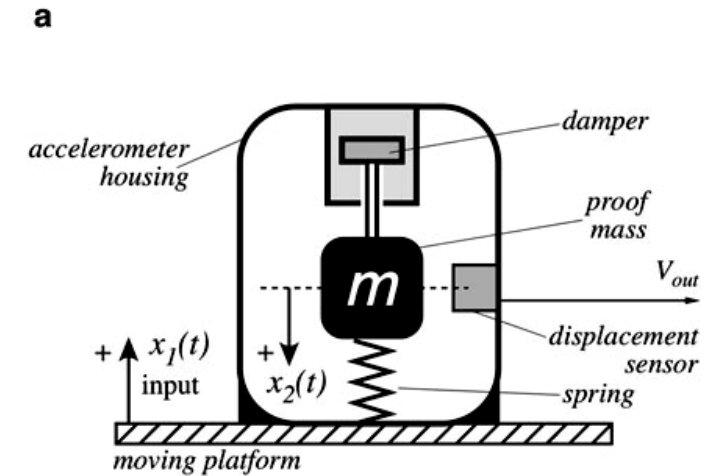


http://www.geekmomprojects.com/wp-content/uploads/2013/03/mems_gyroscope.jpg?resize=300%2C240

Inertial Sensors

Accelerometer measure their **acceleration** in space

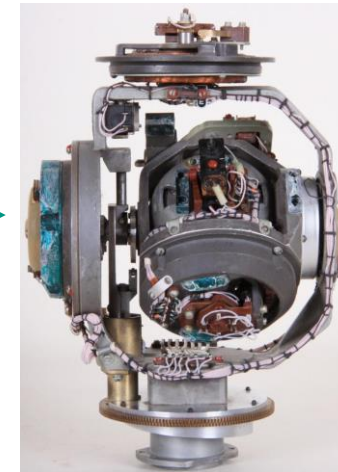
- Applications:
 - Vibration measurement
 - Crash-detection (airbags)
 - Touch detection
 - Head-crash avoidance for falling external hard drives
 -



From Fraden, J.: Handbook of Modern Sensors

Gyroscopes measure their **rotational velocity** in space

- Applications:
 - Navigation (rockets, planes, submarines, ...)
 - Image stabilization (e.g. in smartphones)
 - ...



https://www.ostron.de/out/pictures/z3/kreisbaugruppe_mig21_03_z3.jpg

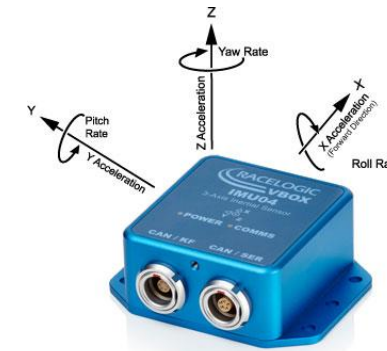
Inertial sensors – (IMUs)

The combination of accelerometers and gyroscopes in one package is called **Inertial Measurement Unit (IMU)**

- Acceleration measurement in all three spatial axes
- Rotational velocity measurement around all three spatial axes
- Often combined with additional magnetometers

IMUs allow for **robust measurement of the absolute orientation** (in the earth's inertial system)

- Requires fusion of different sensor modalities
- Different filters are commonly used for this fusion (Kalman, complementary, ...)



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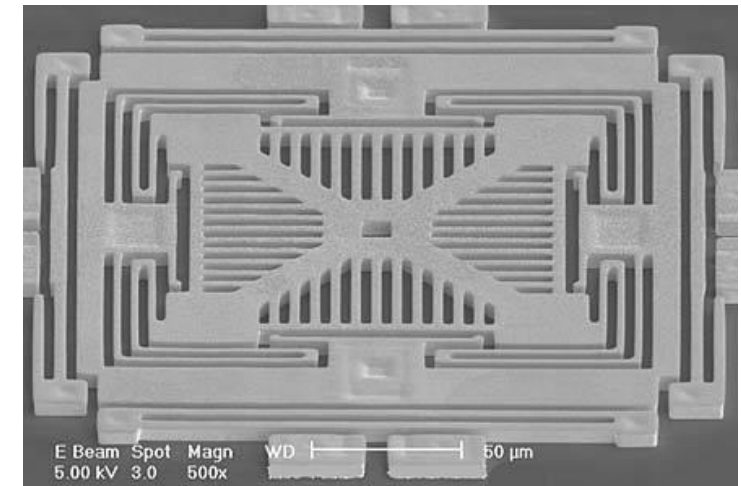
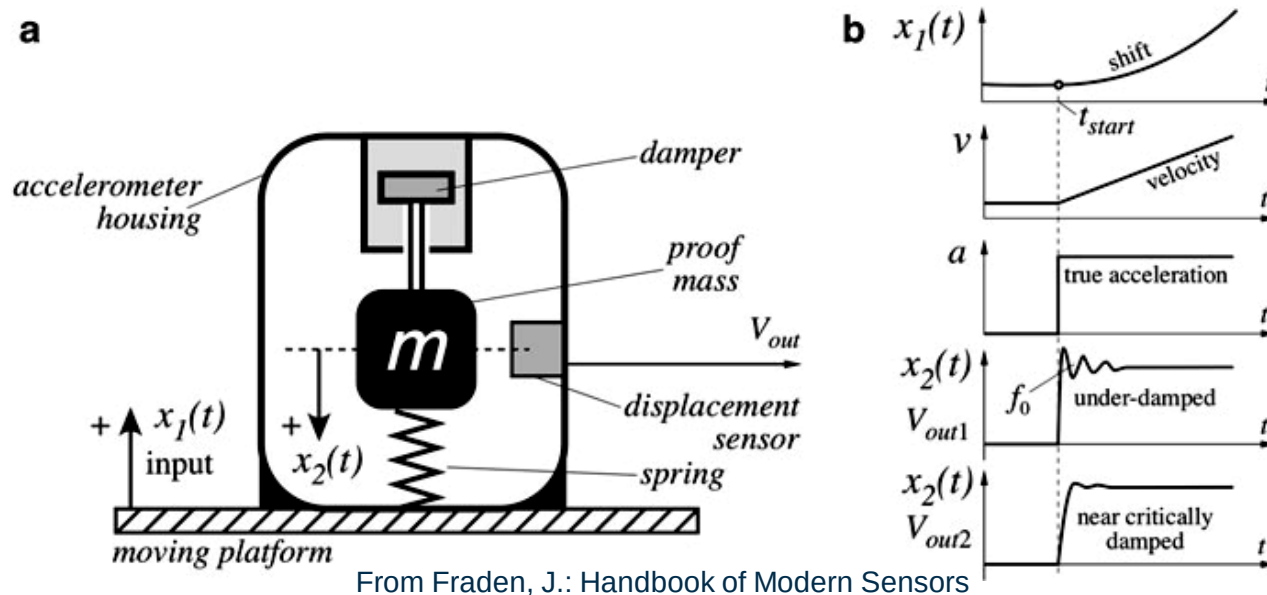
<https://oscarliang.com/ctt/uploads/2013/06/Quadcopter-flying.jpg>

Accelerometers

Accelerometers

Underlying measurement principle: Detect the effects of acceleration on a **seismic proof mass**

- **Most common:** Deflection measurement on a **spring-mass system**
- Accelerometers come in a wide range of scales and use various implementations of the spring-mass-system



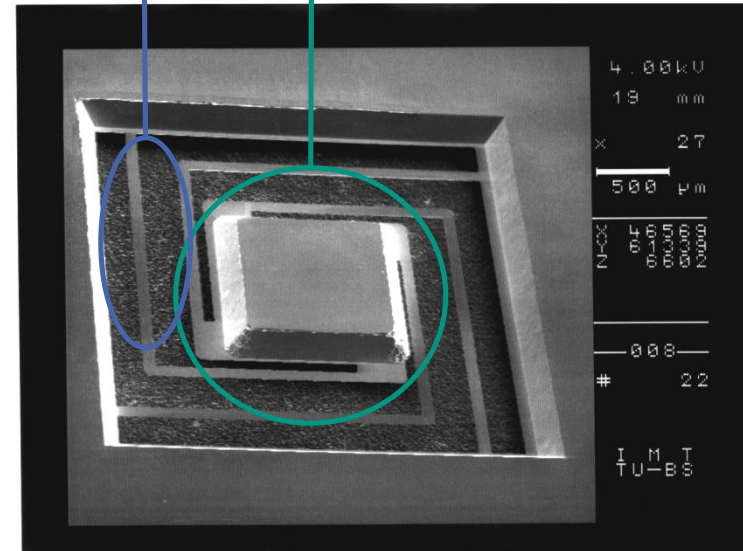
<http://secondelmb.free.fr/edc2/activites/act3.html>

Spring-mass systems for acceleration measurement on different scales and measurement principles

Piezoresistive Accelerometers

Principle of operation:

- Acceleration causes force on a seismic proof mass
- The generated force deforms a **piezoresistive** sensor element
- Due to the piezoresistive effect, the sensor element changes its **resistance** in accordance to the force
- A Wheatstone bridge is used to measure the deformation as in strain gauges



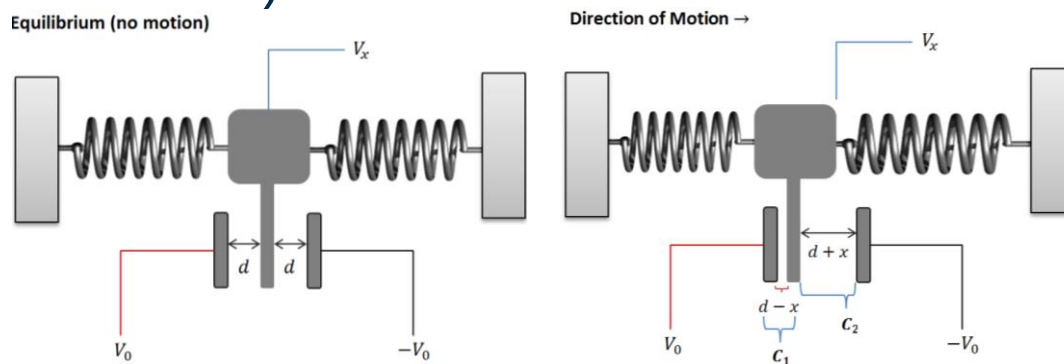
Micro mechanical piezoresistive accelerometer made from silicon

Capacitive Accelerometers

- The seismic proof mass is part of a **capacitor**
- A displacement causes a change in the capacitor's capacity
- The change in capacity can be measured and converted into an acceleration

Differential capacitor

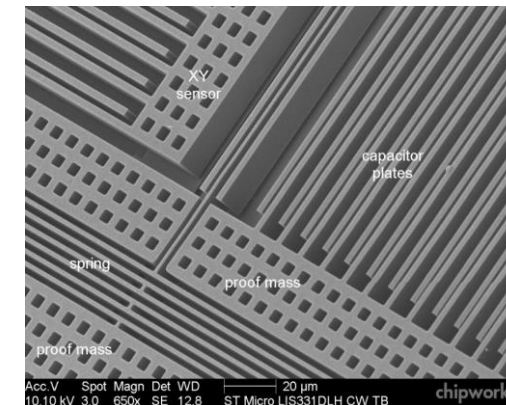
- The seismic mass is (part of) the middle electrode
- Measured using an AC bridge circuit (not in this lecture)



<https://makersportal.com/blog/2017/9/25/accelerometer-on-an-elevator>

Plate capacitor

- The seismic mass is (part of) the plate
- Change in capacity due to the displacement of the plate



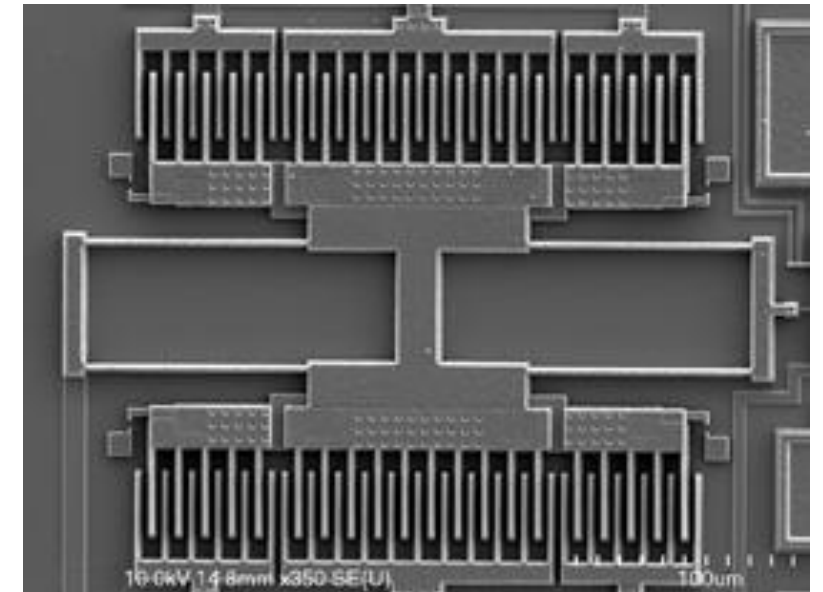
<https://www.rs-online.com/designspark/its-a-small-world-after-all>

MEMS Accelerometers (capacitive)

- MEMS = **Micro Electro Mechanical System**
- Most commonly manufactured using stereo lithography and etching (sensor sizes on the micrometer scale)
- Most common accelerometer type (by far!)
 - Mobile phones
 - Cameras
 - ...

Principle of operation

- Elastically suspended „combs“ form a plate capacitor
- Change in capacity due to deformation of the comb structure



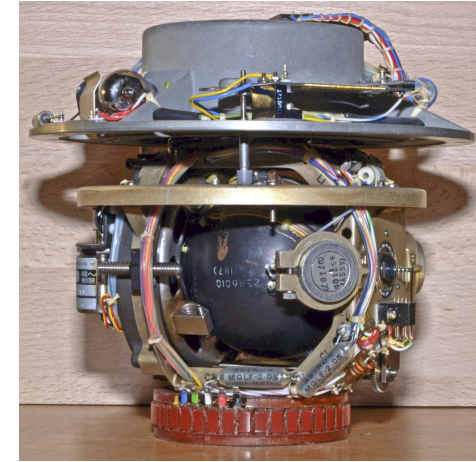
<http://bilderlustige.bid/mems-accelerometer-principle.html>

Micro-mechanical single-axis
accelerometer made from silicon

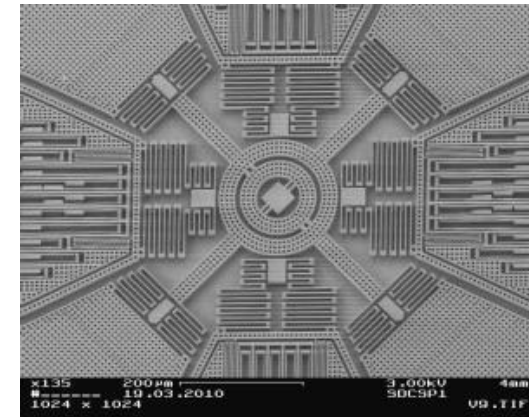
Mechanical Gyroscopes

Use the **Coriolis force** on a moving proof mass to detect their rate of rotation in the inertial system

- Implemented as **spinning gyroscope** (macro-mechanical)
 - In earlier times used for navigation and attitude control of planes, submarines, rockets, ...
 - Rarely used in robotics
- Implemented as **vibrating element** (MEMS)
 - Very small (μm) and very cheap
 - Mass production (smart phones)
 - Most common gyroscope found in robotic applications

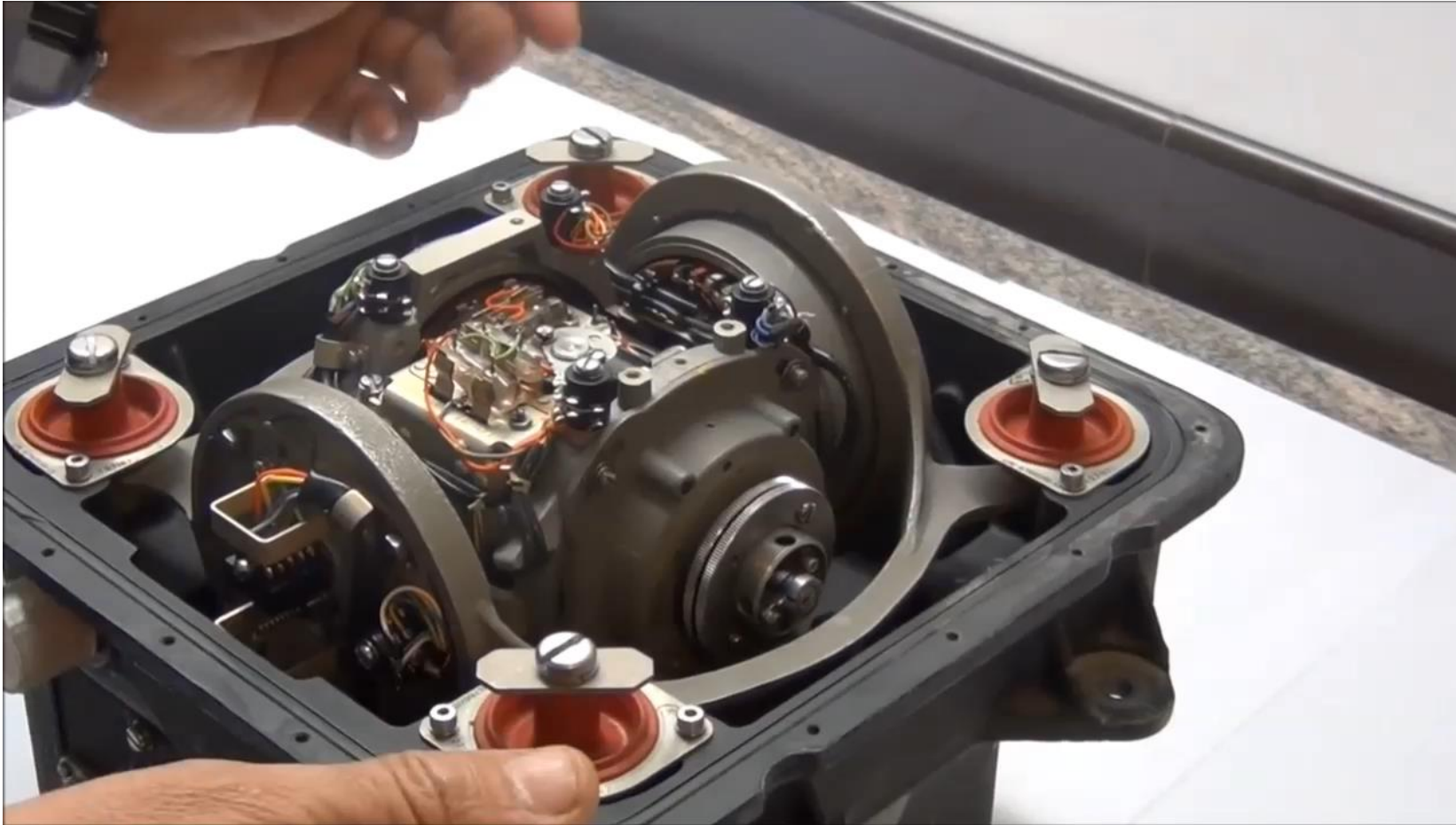


https://de.wikipedia.org/wiki/Kreiselinstrument#/Datei:Gyroscope_hg.jpg



http://www.geekmomprojects.com/wp-content/uploads/2013/03/mems_gyroscope.jpg?resize=300%2C240

Macro-Mechanical Gyroscope – Demonstration



<https://www.youtube.com/watch?v=VycrS3VYjeM>

Macro-Mechanical Gyroscopes – Spinning Gyros

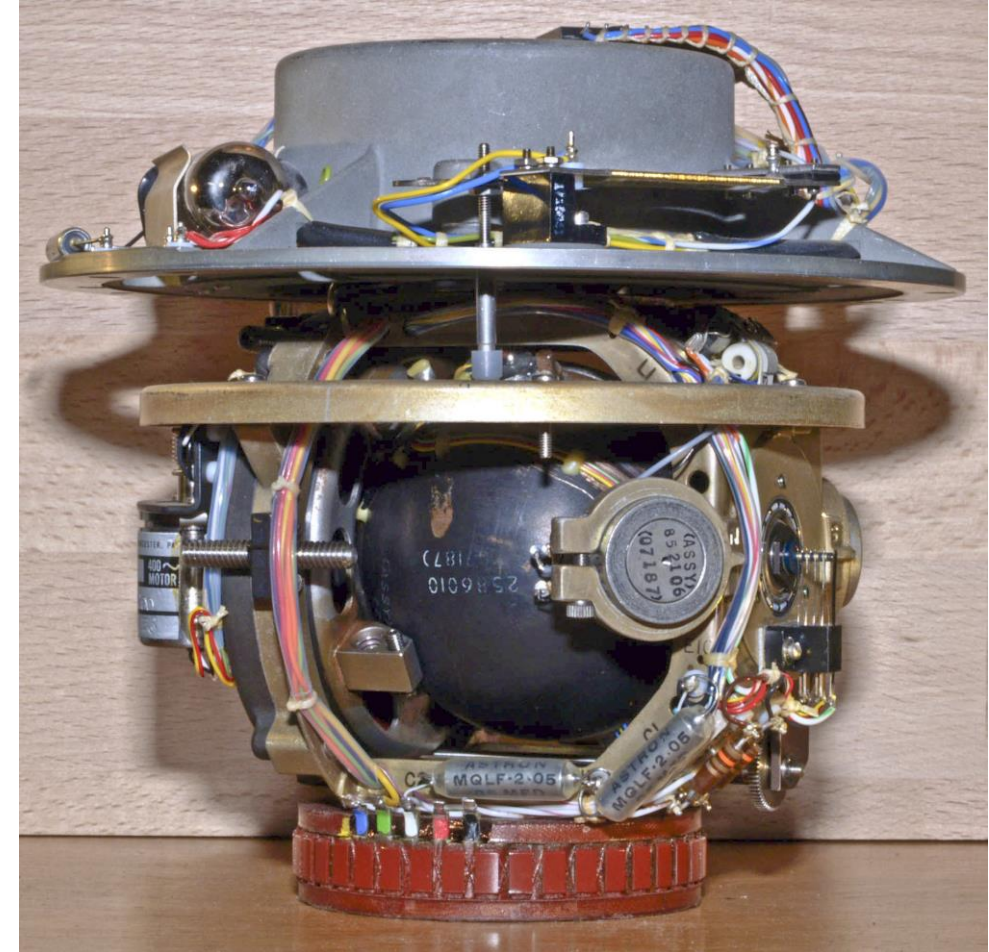
Advantages

- Directly measure the **orientation**, not the angular velocity

Disadvantages

- Mechanically very complex
- High maintenance
- Very expensive
- Large installation space
- Drift caused by friction or asymmetries of the spinner

→ Rarely used nowadays

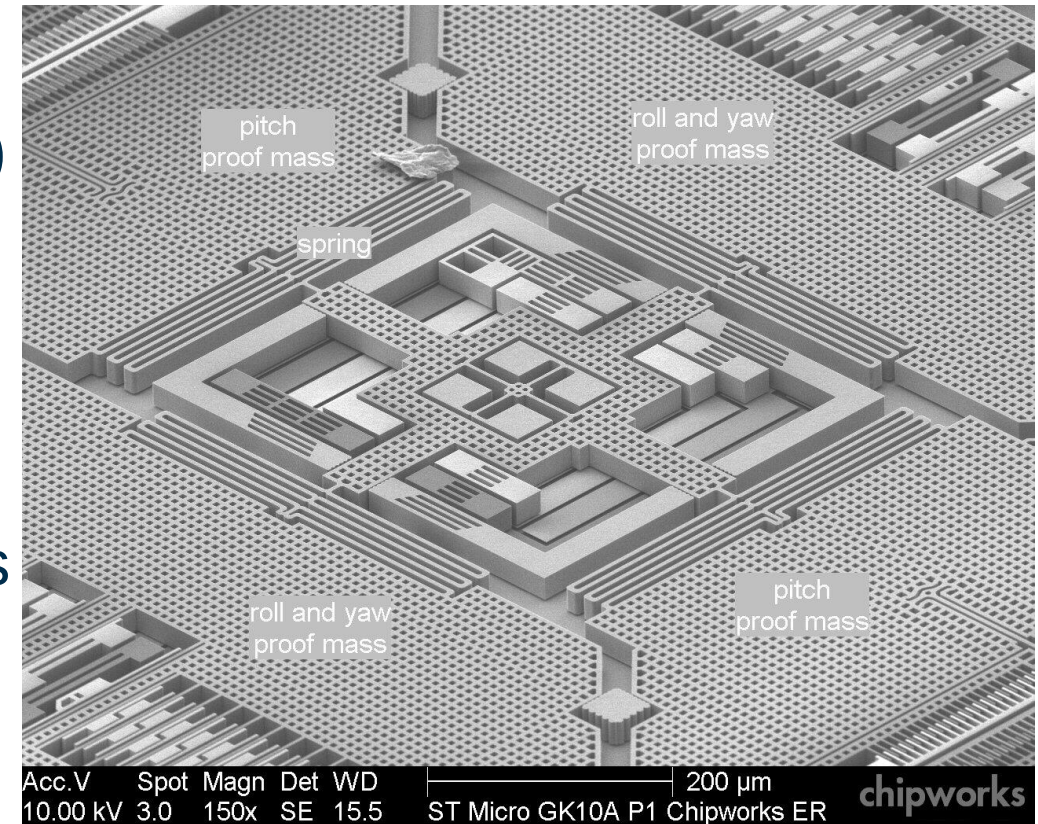


https://de.wikipedia.org/wiki/Kreiselinstrument#/media/Datei:Gyroscope_hg.jpg

Mechanical Gyroscopes – MEMS

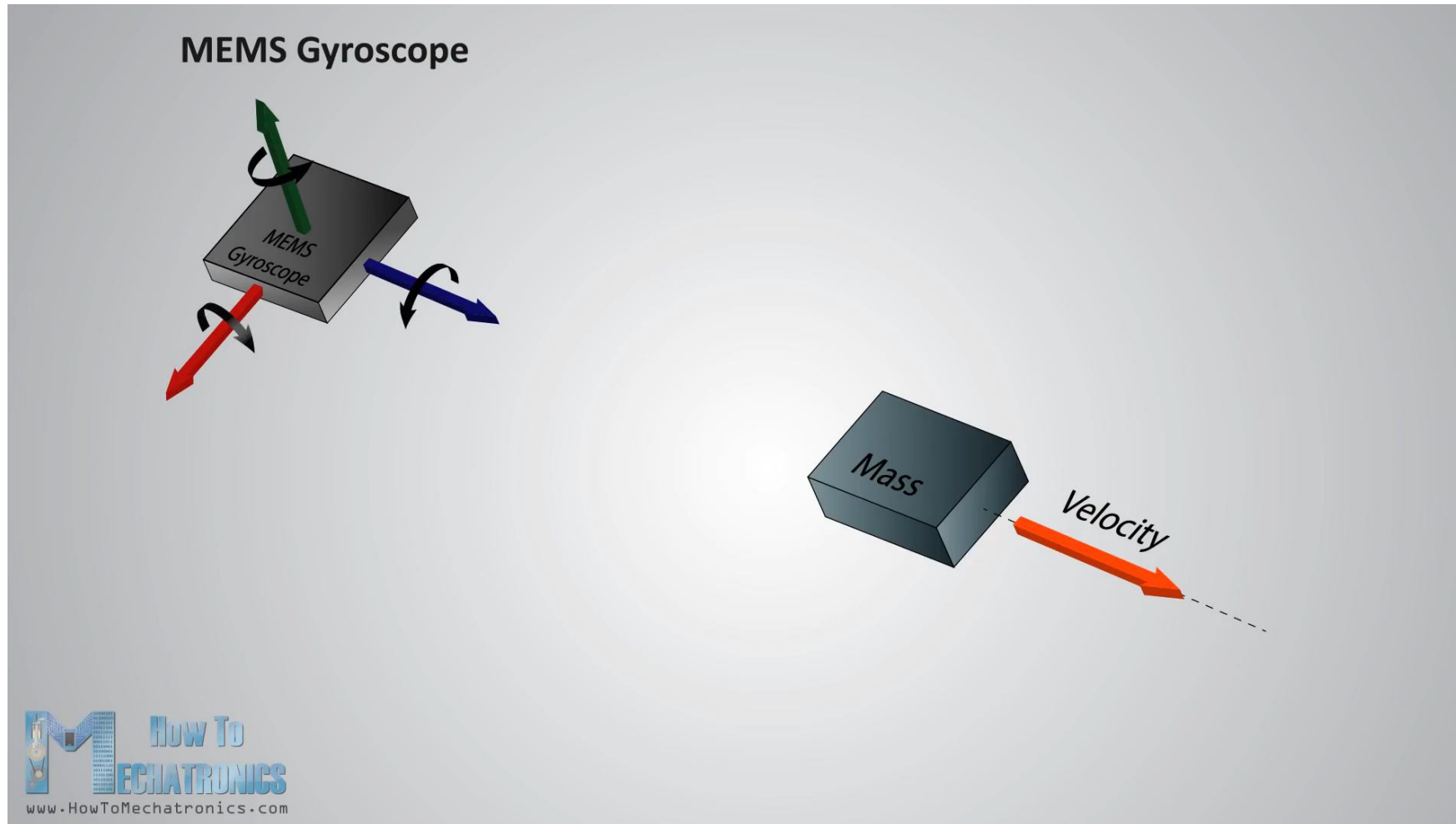
Micromechanical vibratory gyroscopes

- A **high frequency vibration** is excited in an elastic micro-structure (**primary oscillation**)
 - Driven either electro-statically or with piezoelectric actuation
- When rotating, the Coriolis acceleration causes a **secondary oscillation**
- The amplitude of the secondary oscillation is measured and can be converted into the rate of rotation
 - Measurement usually capacitive



Micromechanical 3-axis rotational rate gyro

Micromechanical Vibratory Gyroscopes



https://www.youtube.com/watch?time_continue=11&v=eqZgxR6eRjo

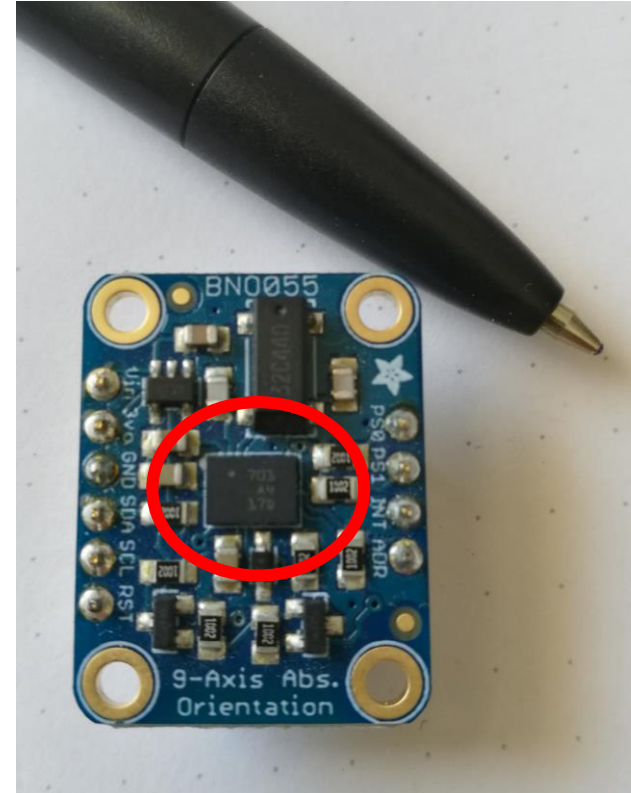
Micromechanical Vibratory Gyroscopes (MEMS)

Advantages

- Very small
- Cheap due to mass production from silicon
- Maintenance free

Disadvantages

- Drift depends on temperature
- Not precise enough for inertial navigation



Modern orientation sensor with (among others) integrated 3-axis gyroscope and 3-axis accelerometer on a single IC

Optical Gyroscopes

- **Use of Sagnac-Effect:** (Georges Sagnac in 1913)

Two light beams traveling in opposite directions around a rotating loop take **different times to complete the circuit**. This time difference leads to a **phase shift** between the beams when they meet again

→ Phase shift is directly **proportional to the angular velocity**

- **Ring laser gyroscope (RLG)**

- Laser beams from a common source go in different directions around a center, **guided by mirrors**

- **Fiber optic gyroscope (FOG)**

- Laser beams from a common source go in different directions around a center within an **optical fibre** (with many windings)



Georges Sagnac

<https://journals.openedition.org/hrc/docannexe/image/5164/img-1.jpg>



RLG

https://de.wikipedia.org/wiki/Laserkreisel#/media/Datei:Ring_laser_gyroscope_at_MAKS-2011_airshow.jpg

Orientation from Gyroscopes

Most commonly the variable to be measured is the rotational rate, not the orientation

- To derive the orientation from a gyroscope, the measured values need to be **numerically integrated over time (dead reckoning)**
- Errors in the measurement are amplified by the integration
 - → **Orientation drift**
- **Due to the integration, the accuracy of the gyroscopic measurement is of very high importance for inertial navigation**

Approximate values of orientation drift using different sensors technologies:

	RLG	FOG	MEMS
Drift [°/h]	0,001 - 10	0,1 - 50	5 - 18000

Gyroscopes - Classification

MEMS

Advantages

- Small
- Cheap
- Easy to integrate

Disadvantages

- Poor accuracy
- High drift

Applications

- Smartphones, cameras, drones, **robotics**,...



FOG

Advantages

- Very precise
- Cheaper than RLG
- Robust

Disadvantages

- More expensive and bigger than MEMS

Applications

- **Robotics**, planes, submarines, rockets



RLG

Advantages

- Extremely precise
- Very little drift

Disadvantages

- Expensive and big
- High technical complexity

Applications

- Military (missiles, submarines, ...)



Summary – Force, Torque and Inertial Measurements

■ Force Sensors

- 1D and 3D Force Sensors or 6D Force/Torque Sensors
- Conversion of physical deformation to digital signal is required (Analogue-to-Digital Converter)
 - Capacitively (MEMS)
 - Resistively (Strain Gauges)

■ Torque Sensors

- Measurement of torsional deformation via strain gauges (analogue) or absolute angular displacement (digital)

■ Inertial Sensors

- Accelerometers measure **acceleration** in space
- Gyroscopes measure **rotational velocity** in space

Inertial Measurement Units (IMU)

The **combination of gyroscopes and accelerometers** is called *Inertial Measurement Unit, IMU*

- Usually both sensor modalities cover **all three spatial axis**, making the IMU a **6D sensor**
- Most common application: 3D orientation measurement

MEMS-based IMUs are mass-produced and can be found in many places in our daily lives

- Smartphones
- Drones
- Game-controller
- Human motion capture
- Robotics (gaze stabilization, balancing)
-



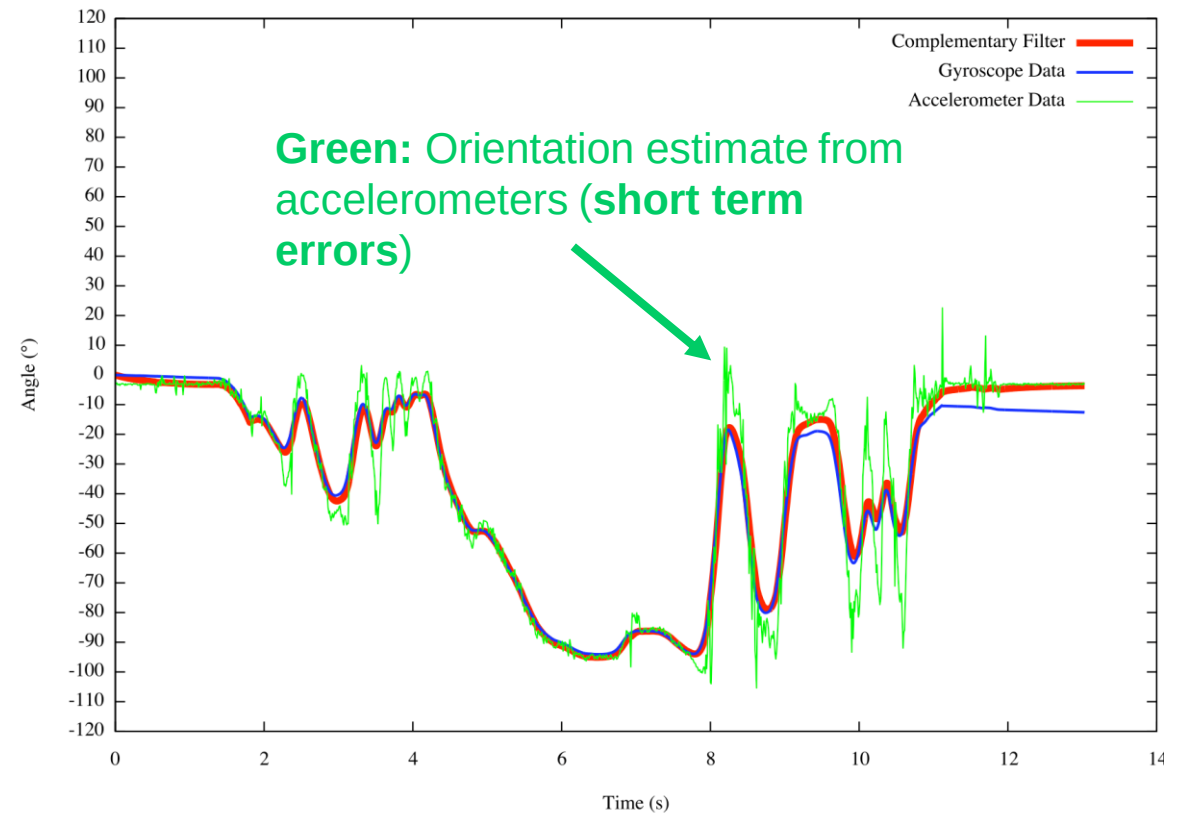
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Orientation Sensing with Accelerometers

In IMUs, the two axes of orientation relative to the horizon (roll, pitch) can be derived from accelerometers

From accelerometers

- The direction of the gravity vector is measured and gives an estimate of the **absolute** orientation (in the absence of motion-induced accelerations)
- **Advantage:** Two axes of orientation are directly obtained, no drift
- **Disadvantage:** (short term) Accelerations that are not due to gravity cause errors in the measurement



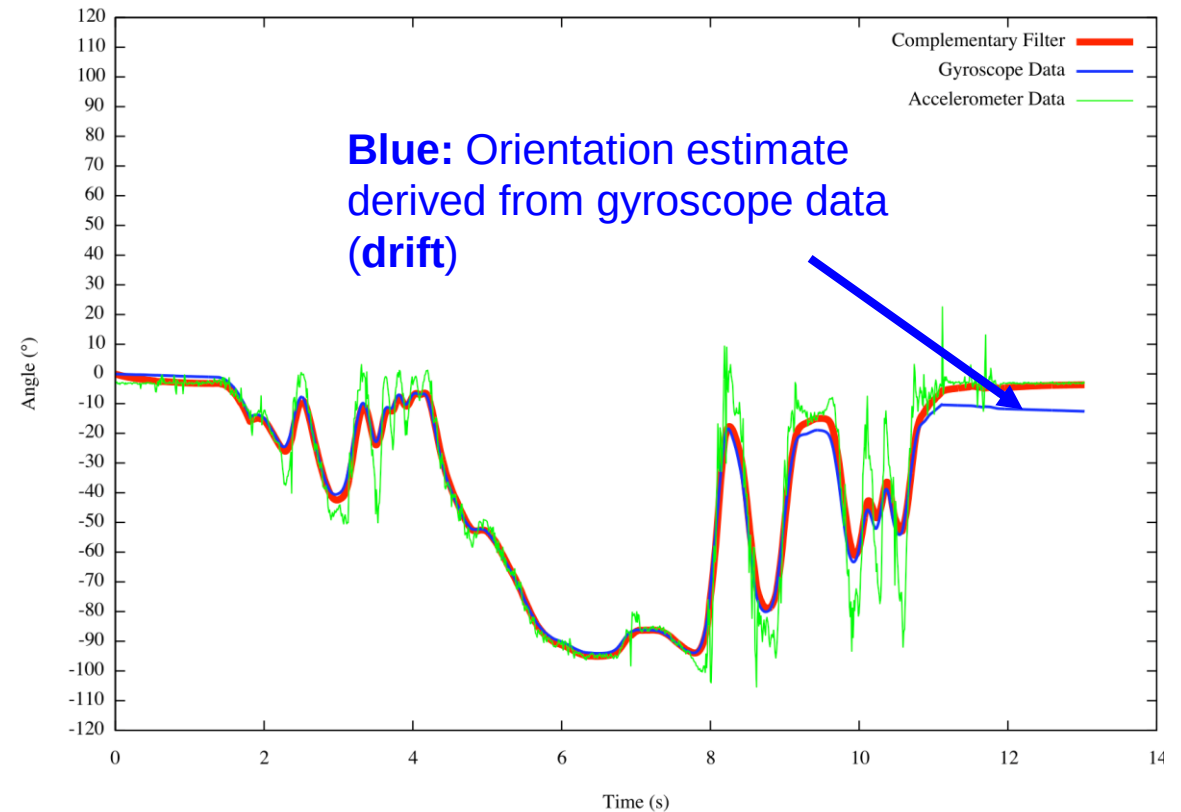
http://www.pieter-jan.com/images/Complementary_Filter.png

Orientation Sensing with Gyroscopes

In IMUs, the measurement of the two axes of orientation relative to the horizon (roll, pitch) can be improved using gyroscopes

From gyroscopes

- The rotational velocities/rates are integrated numerically and provide an estimate of the orientation **relative** to the **initial orientation**
- **Advantage:** Measurement is not affected by linear accelerations (motions) and does not show short-term spikes
- **Disadvantage:** Only **relative** orientation w.r.t. initial orientation; drift caused by the numerical integration cannot be avoided



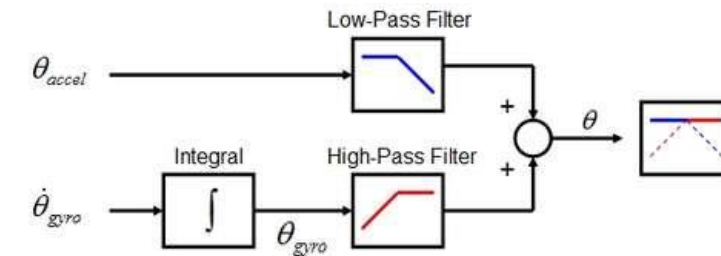
http://www.pieter-jan.com/images/Complementary_Filter.png

Sensor Fusion in IMUs

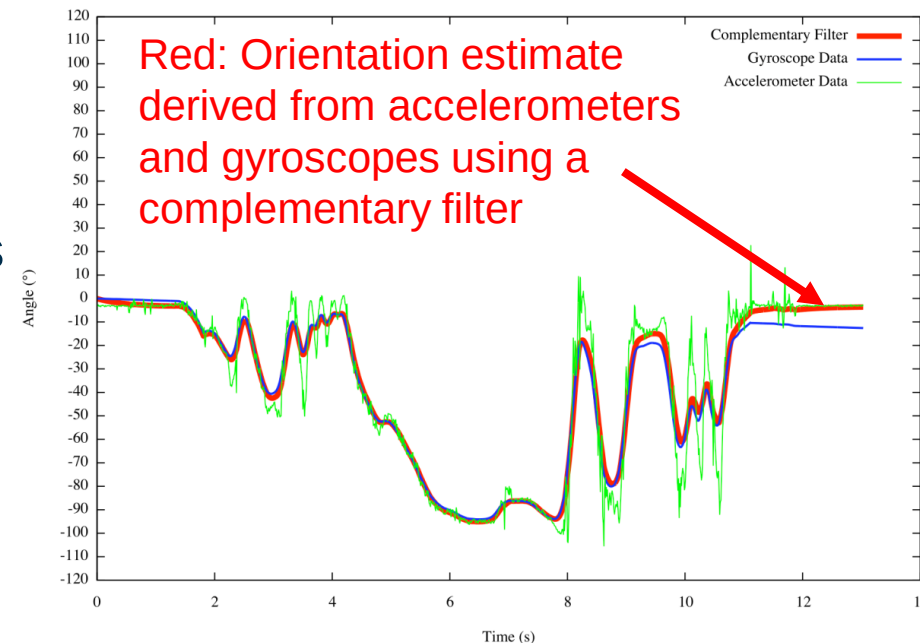
- To leverage the advantages of both sensor types, **fusion** of the sensor modalities is needed
- Different **filter algorithms**
- Common methods:
 - Kalman filter
 - **Complementary filter (simpler)**

Example (complementary filter)

- The plot shows the fusion of the angle estimates derived from both sensor modalities
- The filtered estimate does not have spikes (thanks to the gyroscope) and does not drift (thanks to the acceleration sensors)



<https://i.stack.imgur.com/GTkB0.jpg>



http://www.pieter-jan.com/images/Complementary_Filter.png

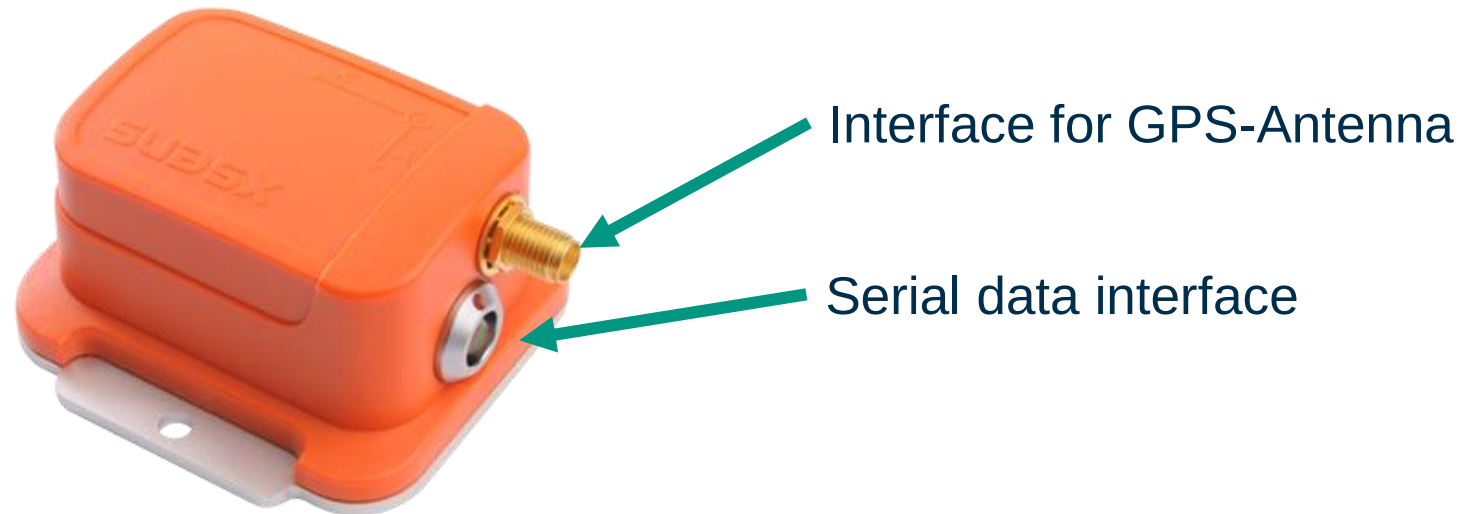
Attitude Heading Reference System (AHRS) - I

AHRS are an extension of IMUs with more sensor modalities and integrated signal processing for advanced orientation sensing

- **Problem when only using inertial sensors:**
 - Orientation about the horizontal axes (*Attitude*: Roll, Pitch) can be determined very well, **but**
 - Drift about the vertical axis (*Heading*) can not be compensated for as gravity does not provide any information there
- **Solution**
 - Addition of more sensor modalities
 - Most commonly a 3-axis magnetometer
 - Provides a **drift-free reference (magnetic north) for the rotation around the vertical axis (yaw)**
- **Result**
 - A **9 DOF sensor** for the drift-free orientation measurement around all three spatial axes

Attitude Heading Reference System (AHRS) - II

- AHRS integrate the signal processing and provide the **computed orientation** as well as the sensor's raw readings
- Other than magnetometers, other sensor modalities can also improve the orientation estimate, above all **GPS** and **barometric pressure sensors**



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MEMS-AHRS with integrated GPS-receiver and barometric pressure sensor

Inertial Navigation Systems (INS)

- *Inertial Navigation System (INS)* provide the orientation and also the global position and velocity with high accuracy
- Consists of an AHRS and possible additional sensor modalities
- Application in (autonomous) airplanes, submarines, land vehicles, missiles, ...



<https://www.youtube.com/watch?v=ymuhJ6pt52o>

Thank you

Next Chapter: External Sensors